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DESIGN OF A FULLY AUTONOMOUS MOBILE PIPELINE EXPLORATION ROBOT
(FAMPER)

A Thesis

Submitted to the Graduate Faculty of the
Louisiana State University and
Basic Science College
in partial fulfillment of the
requirements for the degree of
Master of Science in System Science

in

The Department of Computer Science

by

Jong-Hoon Kim

B.E., Seoul National University of Technology in Seoul, South Korea, 2005
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ABSTRACT

Pipelines have been an integral part of our constructions for many centuries. However, need to be maintained, and the cost of maintenance continues to increase. Robots have been considered as an attractive alternative, and many different types of pipeline robots have been proposed in the past. Unfortunately many of them work under only very restricted environments such as customized pipelines, often have no vertical mobility, or can traverse through only a simple pipeline structure due to wired control.

This thesis presents the design and implementation of a robot based on novel idea we call “caterpillar navigational mechanism”. A *Fully Autonomous Mobile Pipeline Exploration Robot* (FAMPER), for exploring pipeline structures autonomously has been built and its performance has been evaluated. We present the design of a robot based on wall-pressed caterpillar type for not only horizontal, but also vertical mobility in pipeline elements such as straight pipelines, elbows and branches, and its autonomous navigational system providing useful information for pipeline maintenance.

FAMPER has been designed for 6 inch sewer pipes, which are predominantly used in urban constructions. The proposed design enables FAMPER to display formidable mobility and controllability in most of the existing structure of pipeline, and provides a spacious body for housing various electronic devices. Specifically, FAMPER is equipped with several sensors, and a high performance processor for autonomous navigation. We have performed experiments to evaluate the effectiveness of our architecture and we present here a discussion of the performed results.

1. INTRODUCTION

1.1 Robot in Real World Domains

An automatically operated machine that replaces human effort was difficult to imagine; in the view of appearance or perform functions in a humanlike manner. By extension, robotic engineering deals with the design, construction, and operation of robots. A robot is a mechanical or virtual artificial agent, which has a brain of its own. In practice, it is usually an electro-mechanical system, which by its appearance or movements conveys a sense that it has intent or agency of its own.

There were more than one million robots in operation worldwide in the first half of 2008, with roughly half in Asia, 32% in Europe, 16% in North America, 1% in Australia and 1% in Africa. Commercial and industrial robots are in widespread use, these robots performed jobs with greater accuracy with no labor cost and more reliable than humans. Robots can be placed into roughly two classifications based on the type of job they do. The first category includes tasks which a robot can do with greater productivity, accuracy, or endurance than humans, and the other category consists of doing dirty, dangerous or dull jobs which humans find undesirable.

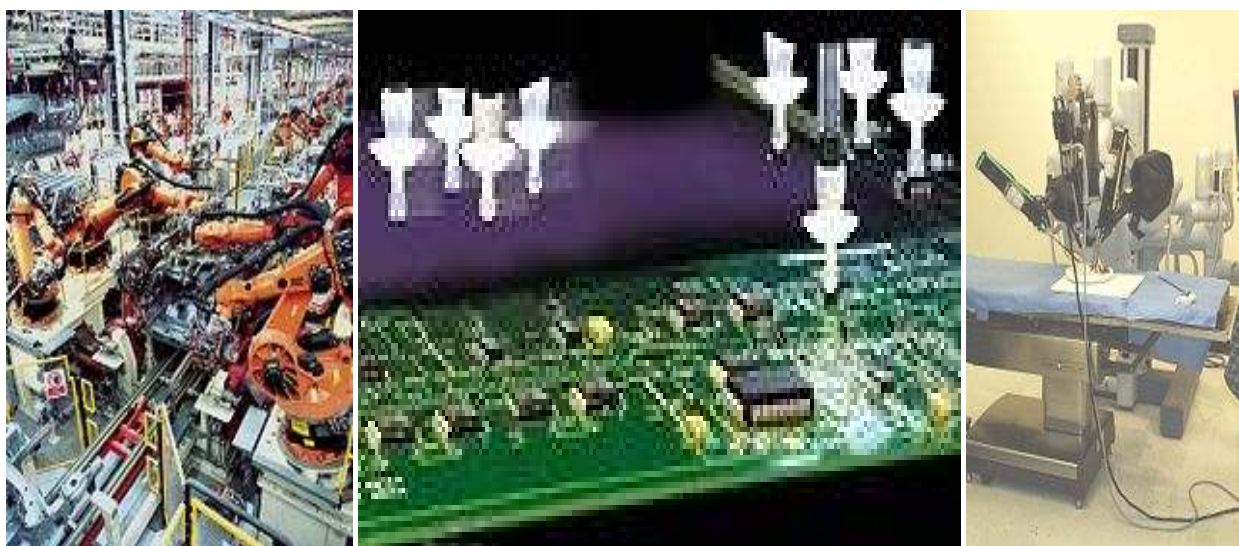


Figure 1. Welding robot in car factory, electronics assembling robot and surgery robot



Figure 2. Floor cleaning robot, landmine removing robot and gas pipeline inspection robot

1.2 The Pipeline Domain

There are many areas where robots can be replaced for human; amongst them pipelines is one of the most challengeable areas. Pipelines have been used in major utilities for long time. Over billions of places from huge plants to an individual house, robots are employed by people. But; many troubles like aging, corrosion, erosion, cracks and physical damages from third parties, have occurred in pipelines. Therefore, maintenance of pipelines is essential in order to keep them functional, and moreover the continuation cost for these activities are being increased. Even with the above mentioned problems in pipeline, people still prefer them. The reason being, pipelines are used in transporting substances through a mere pipe. Most of the time liquid and gases are sent through pipes. Pneumatic tubes that transport solid capsules using compressed air are also being used. Like gases and liquids, any chemically stable substance can be sent through a pipeline. Hence sewage, slurry, water, and even beer pipelines exist. With this knowledge we can classify pipelines with respect to the substance that it carries. We will look at each one in detail.

1.2.1 Oil Pipelines

Dmitri Mendeleev in 1893 suggested pipelines for transporting Petroleum; most countries have employed these pipelines. These pipes started to get widely used around the world. In the year 2007, the total length of oil and gas pipelines in world was almost two millions km, and in the United States had 793,285 km oil/gas pipelines. Pipelines are generally the most economical way to transport large quantities of oil or natural gas over land. Compared to railroad, they have lower cost per unit with higher capacity.

The material used in manufacturing Oil pipes are from steel or plastic tubes with inner diameter typically varying from 4 to 48 inches. Most pipelines are buried underground at a typical depth of about 3 to 6 feet. The oil is kept in motion by pump stations along the pipeline, and usually flows at a speed of



Figure 3. Oil pipeline in Alaska

about 1 to 6 m/s. Multi-product pipelines are used to transport two or more different products in sequence on the same pipeline. Usually in multi-product pipelines there is no physical separation between the different products. Some mixing of adjacent products occurs, producing interface. This interface is removed from the pipeline at receiving facilities and segregated to prevent contamination.

Oil contains varying amounts of wax, or paraffin. In colder climates wax accumulation may occur within a pipeline. Often these pipelines are inspected and cleaned using pipeline inspection gauges. There are various gauges available like PIGS also known as SCRAPERS. These devices are launched from PIG-launcher stations and travel through the pipeline to be

received at any other station down-stream; Cleaning wax deposits and material that may have accumulated along the line.



Figure 4. PIGS and SCRAPERS for oil pipeline inspection

1.2.2 Ethanol Pipelines

These pipelines are majorly used in Brazil and United States. There are several ethanol pipeline projects in Brazil and the United States. Main problems related to the shipment of ethanol by pipeline are its high oxygen content, which makes it corrosive, and absorption of water and impurities in pipelines. Williams conducted an ethanol test in early 1980's. Before the test was conducted; PIGS were used in the pipeline. After the test a few suggestions were made like; frequently dewatering of mainlines using PIGS and spheres, and using closed floater storage tank to prevent rainwater ingestion.

1.2.3 Hydrogen Pipelines

The most cost-effective way to move gaseous hydrogen over a long distance is via pipeline. Hydrogen pipeline is used for transportation of hydrogen through a pipe as part of the hydrogen infrastructure. Hydrogen pipeline is used to connect the point of hydrogen production or delivery of hydrogen with the point of demand, with transport costs similar to compressed natural gas (CNG). Most hydrogen is produced at the place of demand with every 50 to 100 miles an

industrial production facility. The 1938 - Rhine-Ruhr 240 km hydrogen pipeline is still in operation. As of 2004 there are 900 miles of low pressure hydrogen pipelines in the USA and 930 miles in Europe, In Hydrogen transportation, pipeline delivery pressures can go up to 700-1,000 psi.

1.2.4 Water Pipelines

This is one of the most used pipelines all around the world and an ancient method as well. The first people to transport water were the Romans to transport large aqueducts water from higher altitudes by building the aqueducts in graduated segments that allowed gravity to simply push the rushing water along until it reached its intended destination. As time passed by hundreds of pipelines were built throughout Europe and elsewhere, and along with flour mills. The ancient Chinese also made use of channels and pipe systems for public works. The infamous Han Dynasty court eunuch Zhang Rang (189 AD) once ordered the engineer Bi Lan to construct a series of square-pallet chain pumps outside the capital city of Luoyang. These chain pumps serviced the imperial palaces and living quarters of the capital city as the water lifted by the chain pumps were brought in by a stoneware pipe system. We will discuss water pipeline in detail as we go on.

1.3 Sewer/Plumbing Pipeline Robot Domain

As we saw that pipelines are useful for transporting water for drinking or irrigation over long distances when it needs to move over hills, or where canals or channels are poor choices due to considerations of evaporation, pollution, or environmental impact. Plumbing derived from the Latin plumbum for lead, is the skilled trade of working with pipes, tubing and plumbing fixtures for drinking water systems and the drainage of waste. Plumbing is a piping system constitutes the form of fluid transportation that is used to provide potable water to their homes

and business and also remove waste in the form of sewage. The plumbing industry is a basic and substantial part of every developed economy, due to the need for clean water and proper collection and transport of wastes.

A building's waste-disposal system has two parts: the drainage system and the venting system. The drainage system, also called traps and drains, comprises pipes leading from various plumbing fixtures to the building drain (indoors) and then the building sewer (outdoors). The building sewer is then connected to a municipal sanitary sewage disposal system. Where connection to a municipal sewage system is not possible, a local, private, code-approved septic system is required. Cesspools and outhouses do not meet health codes.

Plumbing drainage and venting systems maintain neutral air pressure in the drains, allowing flow of water and sewage down drains and through waste pipes by gravity. As such, it is critical that a downward slope be maintained throughout. In relatively rare situations, a downward slope out of a building to the sewer cannot be created, and a special collection pit and grinding lift 'sewage ejector' pump are needed. By comparison, potable water supply systems operate under pressure to distribute water up through buildings.

Water systems of ancient times relied on gravity for the supply of water, using pipes or channels usually made of clay, lead or stone. Present-day water-supply systems use a network of high-pressure pumps, and pipes are now made of copper, brass, plastic, steel, or other nontoxic material. Present-day drain and vent lines are made of plastic, steel, cast-iron, and lead. Lead is not used in modern water-supply piping due to its toxicity. The 'straight' sections of plumbing systems are of pipe or tube. A pipe is typically formed via casting or welding, where a tube is made through extrusion. Pipe normally has thicker walls and may be threaded or welded, where tubing is thinner-walled and requires special joining techniques such as 'compression-fitting',

'brazing', 'crimping', or for plastics, 'solvent welding'.

1.4 Navigations within Pipelines

As mentioned above, pipelines have to be well maintained for sustaining their functionality, although their material and structure engineering have remarkably improved for the durability. Moreover, cost of the maintenance has been increased tremendously due to increased length of pipelines. In addition, most of sewer pipelines are buried or hided into walls for their protection and hiding their present in the surrounding. Thus, the pipeline accessibility to human for maintenance activities

is very limited because many of them are too small to work for human. Even though there are few big once, people don't want to work inside because of dirty as well as hazardous. For example,



Figure 5. Man unclogging sewer overflow, Chennai, India

sewerage water can be overflowed when sewer pipelines are blocked by sludgy or dirty things. In this case, blockages have to be removed from the pipe. Otherwise all areas can be spoiled by dirty water due to outflow. There are several ways to remove these things. First, blocked pipes can be penetrated with a long stick or wire but it is very difficult to do so when pipelines are bent. Second, blockages can be blown out using air pressure, but it doesn't work when pipe have outlets or clacks between the blockage and starting point of air pressure. Third is an excavation of the area which is suspected for clogging. The difficulty here is a finding clogged area and it also takes long time and large cost. Now we can consider a pipeline robot at this situation. If a pipeline robot can travel, find and remove these things in pipelines, we can significantly reduce

the recovery cost as well as time. We can avoid man power for these jobs. Therefore, a pipeline robot can be a strongly recommendable solution for pipeline maintenance.

After introduction chapter, background is presented in chapter 3. Then FAMPER's concept and design is addressed in chapter 4. Implementation based on this is described in chapter 5 and next experimental results are demonstrated in chapter 6. Finally, conclusion is explained in chapter 7.

2. BACKGROUND

2.1 Mechanical Classifications

A pipeline exploration robot can be broadly classified into two types namely in-pipe and out-pipe. We can clearly perceive that the out-pipe robots are a little less flexible than the in-pipe robots. Also for the conditions which are being considered in the challenges mentioned above, an out-pipe robot would be an in-appropriate choice, as the prime concentration of the FAMPER is to deal with underground or in-wall conditions. So, the current robot “FAMPER” can be classified as an in-pipe robot. Having said that, let us see how the in-pipe robots can be classified into different sub-categories.

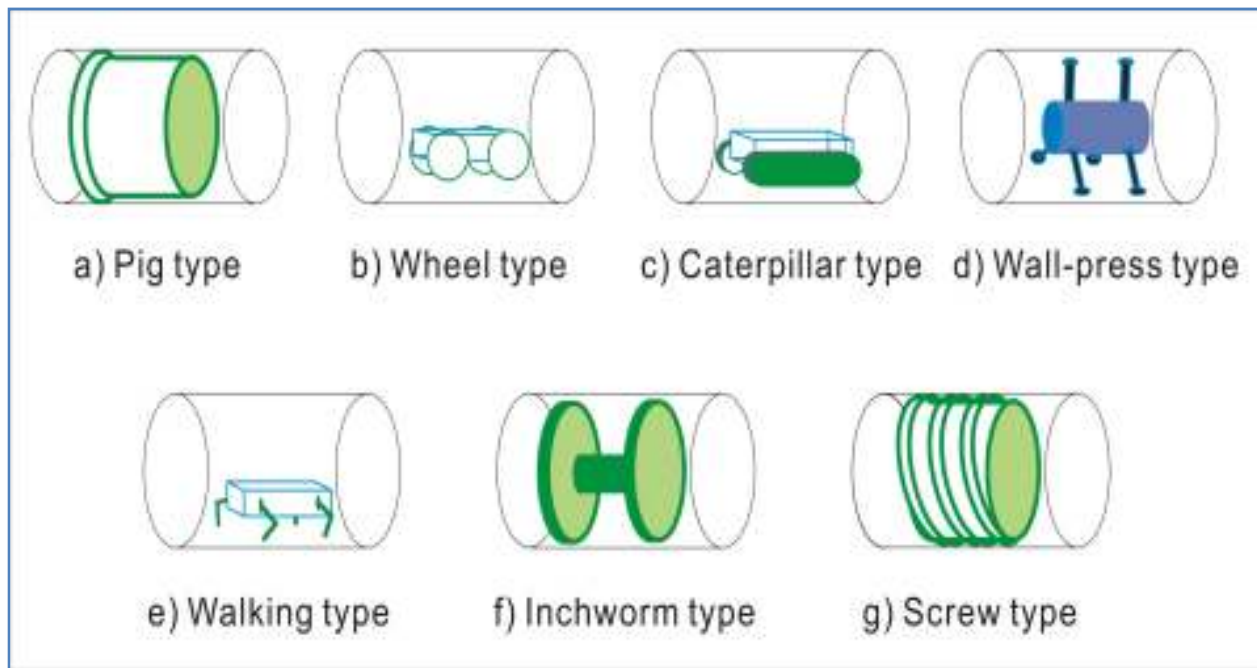


Figure 6. Mechanical classification of pipeline robots. [5]

With a considerable history behind the development of robotics, in-pipe robots can be coarsely classified into seven different sub-categories, based on their applications. These are named as pig type robot, wheel type robot, caterpillar type robot, wall-press robot, walking type robot, inchworm type robot and screw type robot. Let us briefly discuss each one of them and their

probable applications.

Pig type robot as shown in fig 1(a) is one of the most popular commercial robots, which don't generally use additional driving utilities to move along the pipeline. This type of robot is usually used when there is a sufficient flow in the pipeline, which can effectively aid and drive the robot in phase with the flow in the pipeline. Practically this type of robot is used in pipelines with large diameters [1]. Some modifications have been proposed to this type of robots by adding a propeller that will basically make the robot cope up with the speed of the flow. Wheel type robot as shown in fig. 1(b) is one of the basic types of robots, which is very much similar to a plain mobile robot. And a considerable number of commercial robots have been reported for implementing this specific classification [2], [3], [4], [5]. This type of robot is only applicable in horizontal pipelines. One of the prime ways we could see on how to improve the present wheel type robot is to add more gripping feature to the present wheel type robot. Apparently widening the wheels a little bit and adding a band over those wheels can do this. This type is of a robot is called Caterpillar type robot [6] as shown in fig. 1(c). Caterpillar type robots are usually used in conditions that demand much more grip on the walls of the pipeline.

Wall-press type robot as shown in fig 1(d) is another type of in-pipe robot finding its prime usage in vertical pipelines, which need some adequate force to be exerted on the walls of the pipeline that will in turn prevent the robot from falling down. The advantages of wall-press type robot correspond to the robot with flexible mechanism for pressing the wall by whatever means they apply with [5], [7]. Walking type robot as depicted in fig. 1(e) is a robot with articulated legs that will help the robot in movements that are highly sophisticated [8], [9]. This type of robot usually has a complex design due to its sophisticated motion nature, so this design not usually employed unless the pipeline where this is subjected to demands it. As shown in fig.

1(f), Inchworm type robot does in a way mimic the movement of a worm. Apparently this kind of motion is slow and prefers smaller diameter pipelines [10], [11]. As the motion is slow we cannot implement this model for pipelines that are longer in distance. Last but not the least Screw type robot also called helical drive type robot as shown in fig. 1(g) is named after the motion of this robot [12].

Now that we have seen all the different types of in-pipe robots we can have a better prejudice on what features should a given in-pipe robot possess in order for it to be efficient for the job defined. At the same time the robot should perform the required task-space specifications for which it has been designed, like exploring the pipeline. And also depending on the pipeline for which the robot is being designed, we can implement multiple classifications of in-pipe robots, such that the final in-pipe robot will efficiently tackle the complex layout of the pipeline.

The pipeline exploration robots existing today can traverse through the horizontal pipelines but only a part of them can work their way out when the pipelines employ some complex layout that avails one or more of elbows (also called L-Shaped bends) and vertical pipelines. And even from those robots which successfully takes care of the previously mentioned layouts, only a few of them will be able to tackle and negotiate the T-Shaped branches.

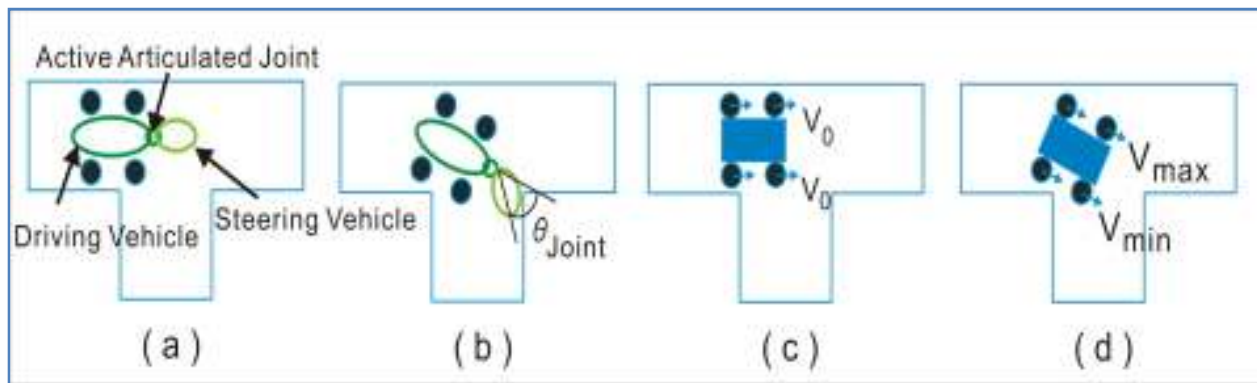


Figure 7. Typical methods of steering in branch. (a) Articulated active joint type: straight drive. (b) Articulated active joint type: steering drive. (c) Differential drive type: straight drive. (d) Differential drive type: steering drive [5]

And a successful implementation of an in-pipe robot demands all these layouts to be handled efficiently, as most of the practical pipeline layouts that exist today employ all these special fittings like elbows, T-shaped branches, vertical pipes, Y-shaped branches etc. in the complete layout of the pipeline structure.

Most of the robots described above also employ some kind of a steering functionality in order to make the robot move. Though they employ specific steering procedures, all of these procedures can be broadly categorized as Articulated-type drive and Differential drive types. The articulated-type robot with active articulated joints is similar to those of a snake or annelid type of reptile, which might be most adequate mechanism, even though the steering mechanism becomes complicated to implement for steering joint [13] and double action universal joint [7]. The alternative way of steering is differential drive steering type as shown in fig. 2 (c) and (d), where speeds of all the driving wheels are modulated in order to steer the robot in those special fittings of the pipeline.

In order to cope up with all the problems and challenges mentioned above FAMPER is designed to employ both the Caterpillar type and Wall-press type, so that it will be able to tackle all those special fittings that are used in the modern pipeline layouts and also increases the vertical mobility and enables the driving modules to change directions in the pipeline.

2.2 Autonomy Based Classifications

Autonomy as the word means by itself leads us to compromise on what different types of sensors, extra hardware and computational equipments we might possibly need to make the robot fully autonomous. Depending on the functionality of the robot, any given robot can be classified as Non-autonomic, Semi Autonomic or Fully autonomic robot.

2.2.1 Non Autonomous Robots

A non-autonomous robot usually just acts as a medium to the human operator to check the subjected area, where the operator cannot reach. The human operator remotely operates the robot, and the control signals for the robot are usually sent through a tethered cable. The human controller determines the conditions of the subjected pipeline by examining the output from the sensor data, which are usually the pictures from the camera attached to the robot. These non-autonomic robots are usually used in commercial plumbing inspection applications.

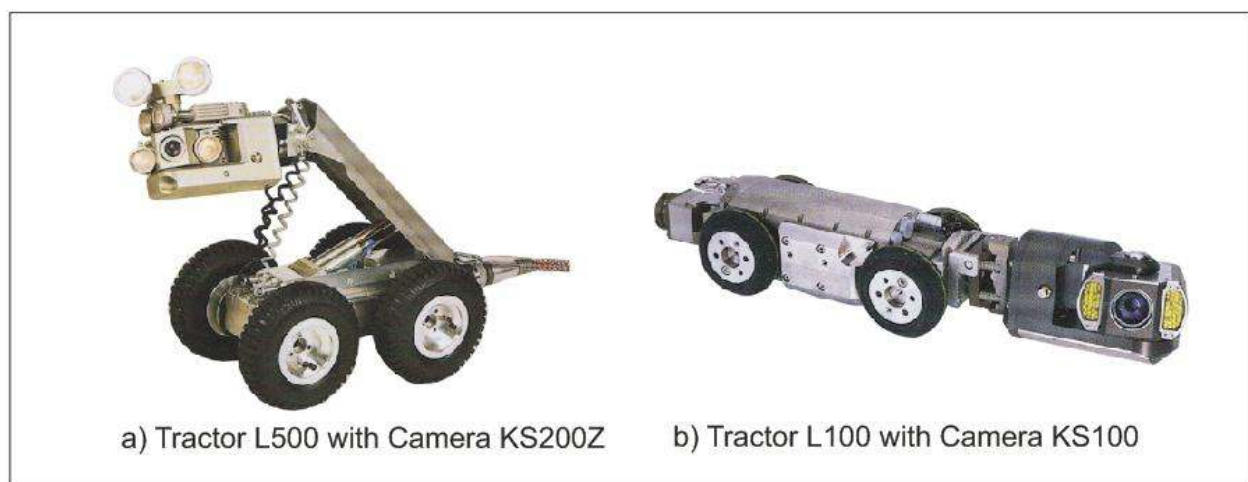


Figure 8. Examples of non autonomous robots; RAUSCH Electronics USA LLC

2.2.2 Semi Autonomous Robots

In semi autonomic robots the assessment of the pipeline is not completely left to the human operator. The Robot often includes modules, which will enable the robot to perform actions, which are usually pre-programmed onto the robotic modules. But still the controls to start these operations have to be issued by the human operator. So this makes the robot a semi autonomic. Some of the robots, which show the semi autonomic functionality, are “PIPAT” [14], [15] developed for quantitative and automatic assessment of the sewage condition. “Pipe Rover/Pear Rover” [16] developed in 1996 for water-filled pipes and ducts can also be

categorized as a semi autonomous robot.

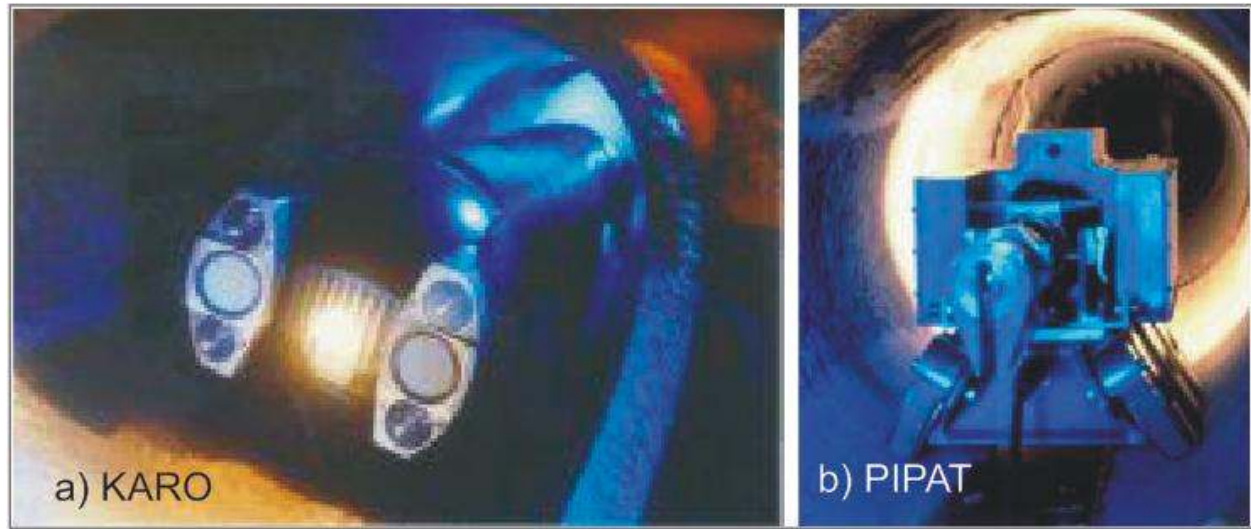


Figure 9. Examples of semi autonomous robot

“KARO” [17] uses a tethered cable as a medium to transmit and receive signals from the controller. It was primarily designed for sewer inspection and testing sensory equipment. So all these robots can be categorized as semi autonomous as they do not have the ability to completely perceive the condition of the pipeline without the prompt intervention from a human controller.

2.2.3 Fully Autonomous Robots

Fully autonomous robots are one such field where the research and development when compared to the other robots is comparatively fewer than the research being done in other types of robots. A Fully autonomous robot usually carries all the required modules that are required for it to assess and process the condition of the pipeline. These are usually un-tethered robots, so all the control signals are transmitted over a radio link. The control programs that usually run on the on-board computing equipment take care of the robotic navigation and the decisions on the paths to be followed by the robot. And time to time all the status messages are communicated to the human inspector or an Artificial Intelligence unit over the radio link, so that if there are any adjustments that are to be made can be communicated to the robot, so that those adjustments will

be adapted by the robot in further actions that will be taken by the robotic controllers. The analysis of the acquired data can be done on the robot itself and/or can be transmitted to the remote inspector for further processing. Only a few of the fully autonomous robots have been developed for pipeline inspection, “KURT” [18] and “MAKRO” are two such robot platforms for pipeline inspection that are designed for fully autonomous navigation in more or less cleaned pipelines with diameters ranging from 300 mm to 600 mm, and under dry weather conditions. KURT is able to navigate to ground level pipe junctions, was designed for inspecting pipelines assisted by maps uploaded into the robot. MAKRO consists of six segments connected by five motor driven active joints, these components enable it to simultaneously climb a step and turn in the pipeline junctions. MAKRO was designed to establish that robots can navigate themselves autonomously inside sewer pipelines.

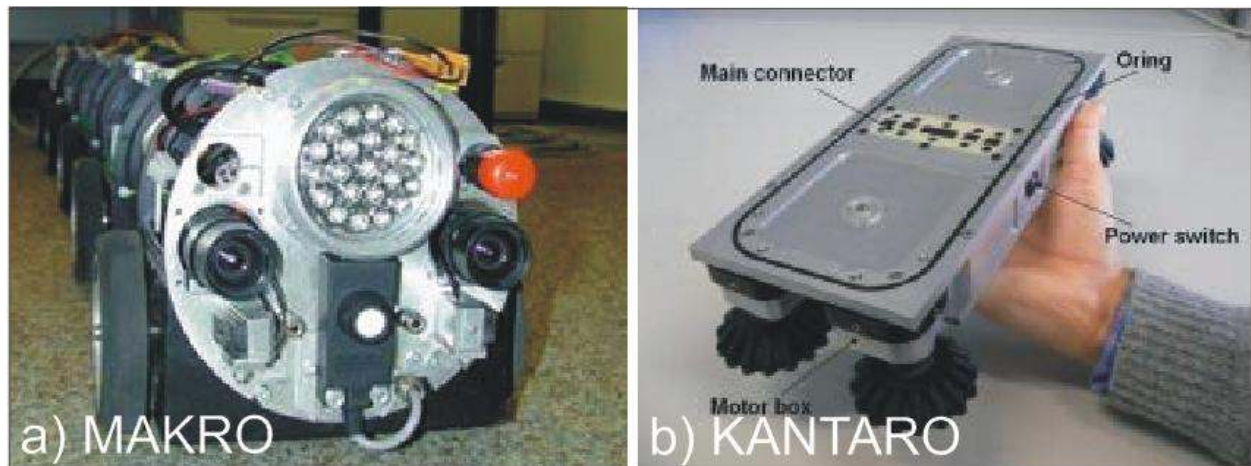


Figure 10. Examples of fully autonomous robots

KANTARO [20] is another fully autonomous robot designed and developed in Japan for navigating through sewer pipelines with varying inner diameter range of 8-12. But this robot only considers the horizontal mobility and the vertical mobility have been ignored.

Although we have quite a few fully autonomous robots for pipelines, none of them guarantee the usage and robustness to be safe and reliable in the pipeline. And most of these

robots include complex navigating mechanism and multiple sensors for motion control. There are a lot of complexities involved in building the navigation mechanism and yet none of the models implemented guarantee the reliability of the system, this makes these robots not to be accepted by the commercial market, although PERMER was more or less successful in implementing a robot for small range of pipes up to 6 inches in diameter.

FAMPER's design is implemented in such a way that it has the ability to be a fully autonomous robot. It has 23 ports for sensors, a high performance processor and big enough memory to conduct all the processing that a pipeline layout demands to be performed on-board. Also FAMPER is equipped with radio link modules, if should there be a requirement that needs the sensor readings to be recorded and notified to a remote inspector. Thus all these features aid FAMPER in becoming a fully autonomous robot.

2.3 Challenges in Pipeline Robot Navigation

2.3.1 Mechanical Challenges

The first and foremost challenge is mobility, a sewer pipeline robot should possess excellent mobility and structure that will enable it to be flexible in horizontal, vertical or any other type of special fitting that is employed in the pipeline layout. One big issue that has to be considered deliberately is the ability to handle the vertical pipeline situations. Vertical pipeline situations also introduce issues related to gravity, frictional force, rolling resistance etc. In addition, layouts including vertical pipelines along with other branching special fittings will complicate these real time issues. Moreover in a given complete pipeline network, we can certainly expect the pipelines to grow and shrink in diameter. So in order to overcome these kinds of situations the robot has to be flexible enough to tackle the challenges. FAMPER is flexible to handle most of these situations, except for being adaptable to variable diameter

pipelines. But FAMPER can be extended in order to include this additional functionality. Most of these real time complications get even more complicated if the subjected pipeline is spoiled or damaged, in some cases this can even make the situations impossible for the robot to solve. However, there are many researches being done and many new approaches are being proposed for reliable accomplishment of mobility, which will always give us better conscience on which particular approach to adapt.

Secondly, another challenge that would occur when dealing with in-pipe sewer pipelines is to make the robot waterproof, because in most of the cases the pipeline is still partially filled with some water that could bring up a potential problem in the operation of the robot. In order to prevent such problems from arising, all the electrical devices should be properly insulated from water.

Thirdly, space management is another issue, which for the most part decides the adaptability of the robot in smaller diameter pipelines. In order to have good and powerful system and additional equipment for increasing their performance, we need spacious body to provide enough room for all these devices. But as the space itself is a limiting feature, we need to find a tradeoff between size and performance.

2.3.2 Electrical Challenges

Firstly as mentioned previously in mechanical challenges section, space issues are to be considered for robot's performance. So, there is always a requirement for devices that are small yet powerful. One such device is Gumstix, which is used in the implementation of the FAMPER. We need to consider the applications and the goals for which the robot has been implemented, while making the selections for the devices on which the robot will be built. For instance, if the robot being designed is a pipeline exploration robot, which does the explorations using sonars,

then considerable amount of space have to be dedicated for housing powerful sonars.

Secondly accurate and reliable control system is a requirement for the robot to detect and respond to the situations encountered in the subjected environment. Wireless connectivity has to be maximized in order for the robot to transmit the sensor data to a remote inspector, so that the processor included on-board can be used in doing things that are proximally demanding. The sensors should be tuned such that the sensor dependent devices do not waste their processing power in doing job for false triggers from the sensors. A good processor that is small yet powerful enough along with large memory will definitely serve an added advantage in the operation of the robot. Also as the robots have to be operated in conditions where providing power becomes impossible without losing mobility, we need to use powerful batteries that could power the robot and the attached components for a period of time. Selecting devices that have low power consumption can increase their longevity reasonable.

2.3.3 Programming Challenges

Though the robot is fully autonomous it should be able to provide the required control interface for the remote inspector in order to assess the performance of the robot. Also the robot should be capable enough to learn and practically implement those learning in real time with the help of Artificial Intelligence based controller software that helps in navigating autonomously.

The Software framework should be chosen in such a way that it would provide the implementation of the software's in multi coordinate system.

2.3.4 Application Challenges

Some of the possible applications for a pipeline-based robot could be, drawing blueprints for pipelines that are under the ground or inside the walls by exploring through the pipeline layout, and simultaneously generating the blueprint. Inspecting the pipelines for any damages

that have to be repaired and fixing them if possible. Cleaning the pipelines is most desirable for maintenance of pipelines. Finding lost items in the pipelines is one of the most pleasing options for general community.

3. DEIGN OF A FULLY AUTONOMOUS MOBILE PIPELINE EXPLORATION ROBOT

FAMPER has three major goals in its concept and design. First and foremost thing is mobility of FAMPER which will enable it to travel in horizontal, vertical conditions involving with any kind of special fittings that a pipeline might be using for a 6 inch pipeline. Secondly the operating architecture being used has to be simple yet flexible enough and adaptable enough for the environment being considered. Thirdly it should be comprised of a powerful computing system for handling navigational complications both autonomously as well as efficiently that pipeline network might be having.

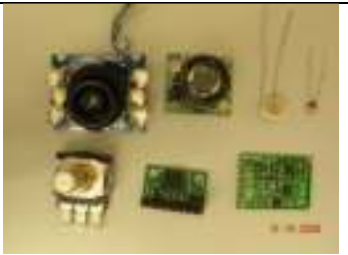
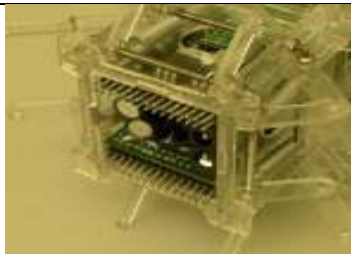
		
1) Wide track belt	2) Independent Suspension	3) Flexible Links
		
4) Spacious Body	5) A bird's eye View	6) High resolution sensors
		
7) Centralized connections	8) Stacked Electrical Board	9) Powerful battery

Figure 11. FAMPER's design feature

3.1 Design Features

FAMPER has four caterpillar tracks that provide good gripping force in both vertical and horizontal pipeline situations. Independent suspensions and links enable FAMPER to travel in any type of pipeline network available. Spacious central body frame allows the ability of installing powerful computing system. Centralized interface provides easily reachable connections to many of the sensors that FAMPER uses. Powerful batteries help FAMPER to be mobile, and also in increasing its ability to perform the given actions long enough in the given mission range.

3.2 The Wall-Press Caterpillar Mechanism

3.2.1 Continuous Track: Caterpillar

The caterpillar tracks used here help the robot to distribute its weight evenly over a larger surface of the track, when compared to the wheel-based robot. The caterpillar tracks have this ability because as the tracked robot moves forward the segment which is used for the track is laid out flat on the ground at the front and picked up again at the back. So this feature of the caterpillar track helps the robot to handle the uneven surfaces much efficiently. The rollers at the front and back of the robot, which aid in the movement of the rolling segment, take up the additional load. The complete rolling track of the robot will help the robot in handling the movement smoothly in loose areas, where the wheel-based robot of the same potential would fail to do the job. Also the caterpillar-based robot will have additional grip provided from the rolling segment in situations with surfaces having steeper gradients and slippery textures. We can see that for equilibrium the ground pressure of a car must be equivalent to the pressure of the air in the tires. For most of the cars this is around 30 psi (207 kPa), whereas for a seventy-ton M1 Abrams tank has the ground pressure of around 15 psi (103 kPa), which is exactly half that of the

car. Thus we can see that the caterpillar track helps the robot to exert less pressure on the pipeline than when compared to the wheel-based robots in a horizontal pipeline. In case of a vertical pipeline a wheel-based wall-press robot have to exert considerably more pressure on the pipeline walls than when compared to the caterpillar-based wall-press robot, because the caterpillar-based wall press robot have larger area in contact with the surface of the pipeline. Also when passing onto any kind of special fittings from the vertical pipeline, a wheel-based wall-press robot might lose its balance when the contact between its wheels and the pipeline surface is lost, but in case of a caterpillar-based wall-press robot the rolling segment will always have contact with the pipeline surface, so it will prevent this robot from losing its balance.

Also as we have discussed before, the track-based robot will have better mobility than the regular wheel-based robots over uneven surfaces, tracks smooth out the bumps and glide over small obstacles. Tracks are less likely to get stuck up on the muddy surfaces as they distribute the weight all over the rolling segment. Tracks can also give better mobility, as a track-based robot can turn in its own radius by driving the tracks in opposite directions.

The main disadvantage with track-based robot is, tracks have lower top speeds due to friction but this friction is necessary to vertical movement in pipeline. Tracks-based robots are mechanically challenging, when compared to wheel-based robots. But caterpillar-based robots cause lesser damage to the inner surface of the pipeline because of their smooth rolling surface, than the wheel-based robots. Thus we can say that the advantages of using a caterpillar-based robot will convince us enough to adapt track-based architecture for FAMPER.

3.2.2 Independently Enlargeable Suspension Link Model

In the mechanism used in the FAMPER, the distance between the central body and the tracks can be determined based on the movement of the flexible links, the elastic restoration

force of the spring at the each suspension link, and reaction forces from the wall. From the above figure.12 we can deduce the height of the track from the central body using the Pythagorean Theorem, and the following equations can be derived,

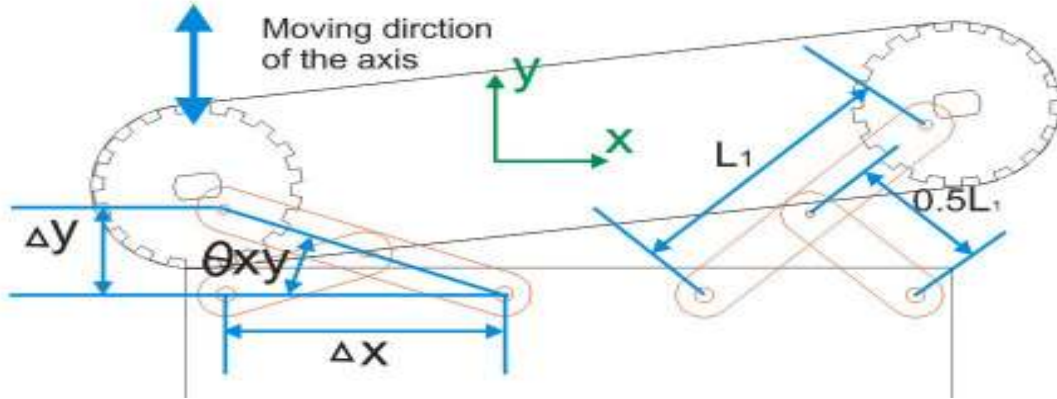


Figure 12. Model and feature of FAMPER's link mechanism.

Here L_1 is the length of the link connecting the track and the body also called crank, another link of length $\frac{1}{2} L_1$ connects the midpoint of the crank and the central body. θ_{xy} is the suspension angle between the crank and the central body of the robot. Δ_x and Δ_y are the displacements along the x and y axes respectively. The vertical displacement of the track from the central body can be calculated using the above equation involving the horizontal displacement Δ_x of the links on the central body. The force with which the tracks press on the

inner surface of the pipeline can be determined by adjusting the stiffness of the spring in the initial stage of designing, and also the tractive force of the robot is determined accordingly. Since the above-mentioned mechanism has been implemented in building FAMPER, we can be sure that FAMPER will have effective contact with the inner surface of the pipeline, and also the adaptable nature of the tracks and suspensions will make sure that FAMPER can sustain even in conditions with uncertain pipeline conditions, as well as providing sufficient traction forces during movements.

$$\tan(\theta) = \frac{\Delta y}{\Delta x}$$

and by Pythagoreans' theorem

$$L1^2 = \Delta y^2 + \Delta x^2$$

$$=> \Delta y^2 = \Delta L1^2 - \Delta x^2$$

$$=> \Delta y = \sqrt{\Delta L1^2 - \Delta x^2}$$

and also

$$\Delta y = \Delta x \tan (\theta_{xy})$$

Also FAMPER is additionally equipped with the capability of having suspension independent of other suspension modules. In other words we can say that the track can move in any particular way, in order to meet the subjected conditions and the pipeline situations. These independent suspensions help the caterpillar track to be more flexible as shown in the pictures below.

When the track meets any kind of obstacles, which only obstructs only a part of the track, the track has the ability to be tilted for increasing the contact surface of the track. This feature of having flexible tracks increases the gripping force of the track, which is very helpful in conditions where pipelines are damaged.



Figure 13. Front view of FAMPER's link system

3.2.3 Caterpillar Vs Other Mechanisms

Pig type is not suitable in sewer pipeline network because it is difficult for it to turn in elbows and branches as well as to move without sufficient driving flow by water or any carrier. Thus, most of pig type robots are used in long and straight pipeline such as oil or gas pipelines. Screw type also faces the difficulty in turning at elbows and branches. Wheel type has been used for many different sewer pipeline robots but most of the sewer pipelines have some kind of dirt on the inner walls that could make the movement of the wheels problematic. When wheel-type robot is combined with wall-press type, it can have vertical mobility but, the exerted force on the walls of the pipeline should be completely coming from the wheels, so if one of the wheels loses contact from the inner pipeline surface, the robot might lose its balance. In general, it has more simple mechanism than caterpillar type but caterpillar with wall-pressed type gives more stable grip because of large surface contact in vertical situations. Walking type robot can have climbing ability but needs complicated mechanical and control systems. Especially, this type is more difficult to make for small sized pipelines. Inchworm type is too slow to travel longer distances. Thus, wall-press caterpillar type is best solution for sewer pipeline inspection robot.

3.3. FAMPER Operational Architecture

FAMPER's operational architecture consists of three operator modules; Supreme Operator, Perception Operator and Action Operator. Each operator has different manager

modules which perform fundamental operations with their own property values. The property values are saved in a central repository which can be referred and updated by the module which is accessing it and/or any other operator/s based on the rules formulated. This modular ability of FAMPER's architecture enables each module to be developed independently, for instance action operator as we know only depends on the property values that are defined for the action operator, now the way the action is performed can be changed only by changing the property values of the action operator. These property values can be changed by supreme operator or perception operator depending on the environmental conditions or the changes that any other operator might require for performing that action.

As the architecture is modular, we can manually set those property values and the decision message in order for the action controller to take the required action. The supreme operator just send the decision message to the action operator and also supreme operator might tune the property values of the action operator to make the action being performed adaptable to the environment, based on the indirect feedback from the perception operator in real time.

These messages can be feedback indirectly and receiver's property values can be tuned by message sender using that feedback. For example, Perception Operator senses 90 degree branch and sends sensor message to the Navigating Manager in Supreme Operator. Supreme Operator makes the decision to turn 90 degree and Navigator manager just sends their decision message to the Caterpillar Manager in the Action Operator, for instance "turn 90 degree" instead of each motor speed value. After receiving the message, the Caterpillar Manager reads their properties about 90 degree turn, and controls the DC motors in caterpillars tracks using different speed values.

When the action event is started, Caterpillar Manager sends the action message to

Perception Operator. The Perception Operator then observes changes and sends the observed result to Supreme Operator within monitoring term property value in central repository. Supreme operator tunes properties of Action Operator based on the sensor message which is a feedback of the decision to be taken. The central repository contains properties of operators and managers, rules, sensed data, map information, results of actions and history of events.

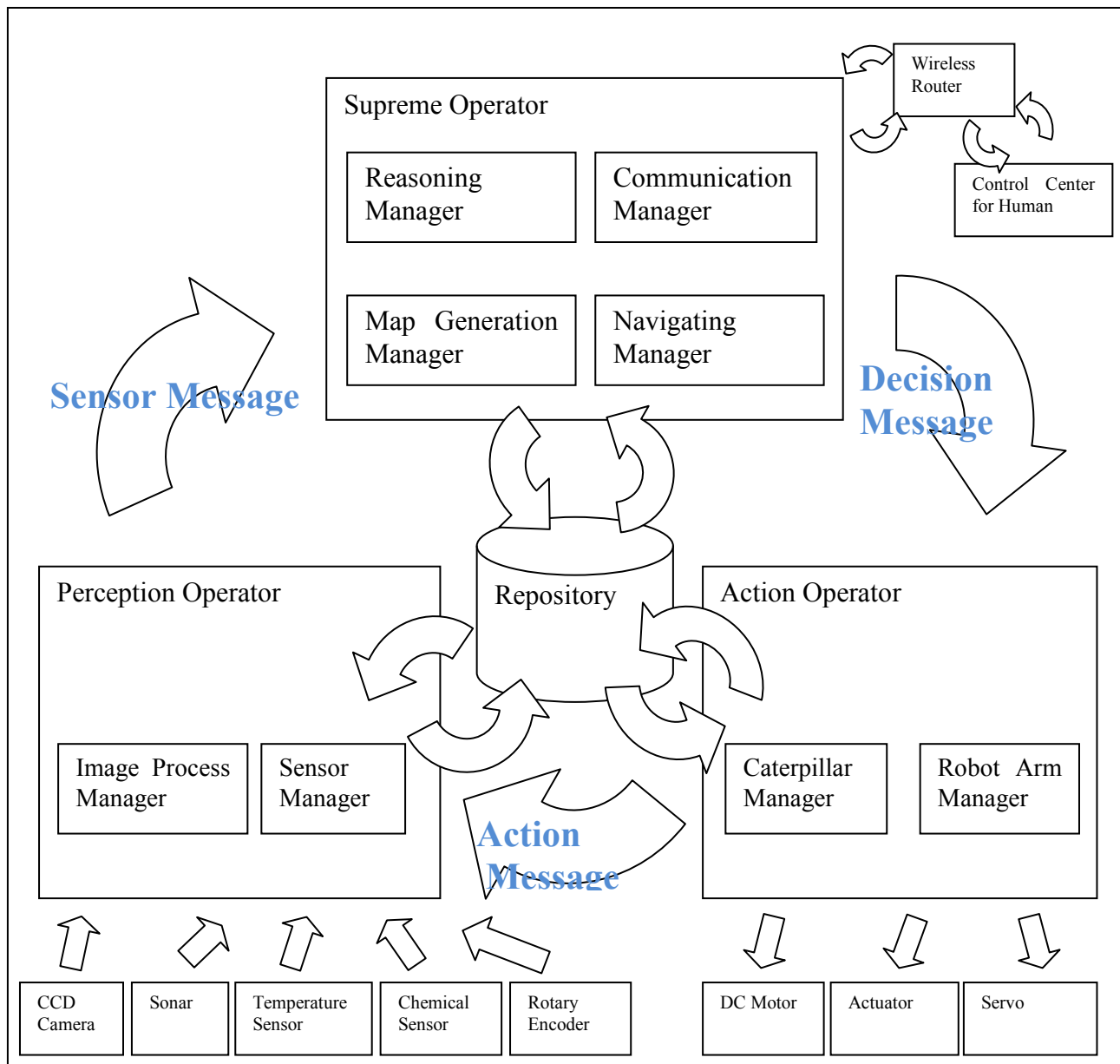


Figure 14. Flow-diagram for FAMPER's operational architecture

3.4 Control Methods for Three-Dimensional Movement in Pipelines

Proposed model enables FAMPER to turn easily at elbows and branches without complicated control mechanism by using four independent caterpillar tracks. With various combinations of speeds and rotating direction of tracks, FAMPER can perform very sophisticated turning operation in any three dimensional direction. Figure 15.describes motor speeds for all the 4 tracks depending on the direction in which robot is intended to turn. If a robot wants to move forward, Z direction, all tracks rotate with same direction and speed.

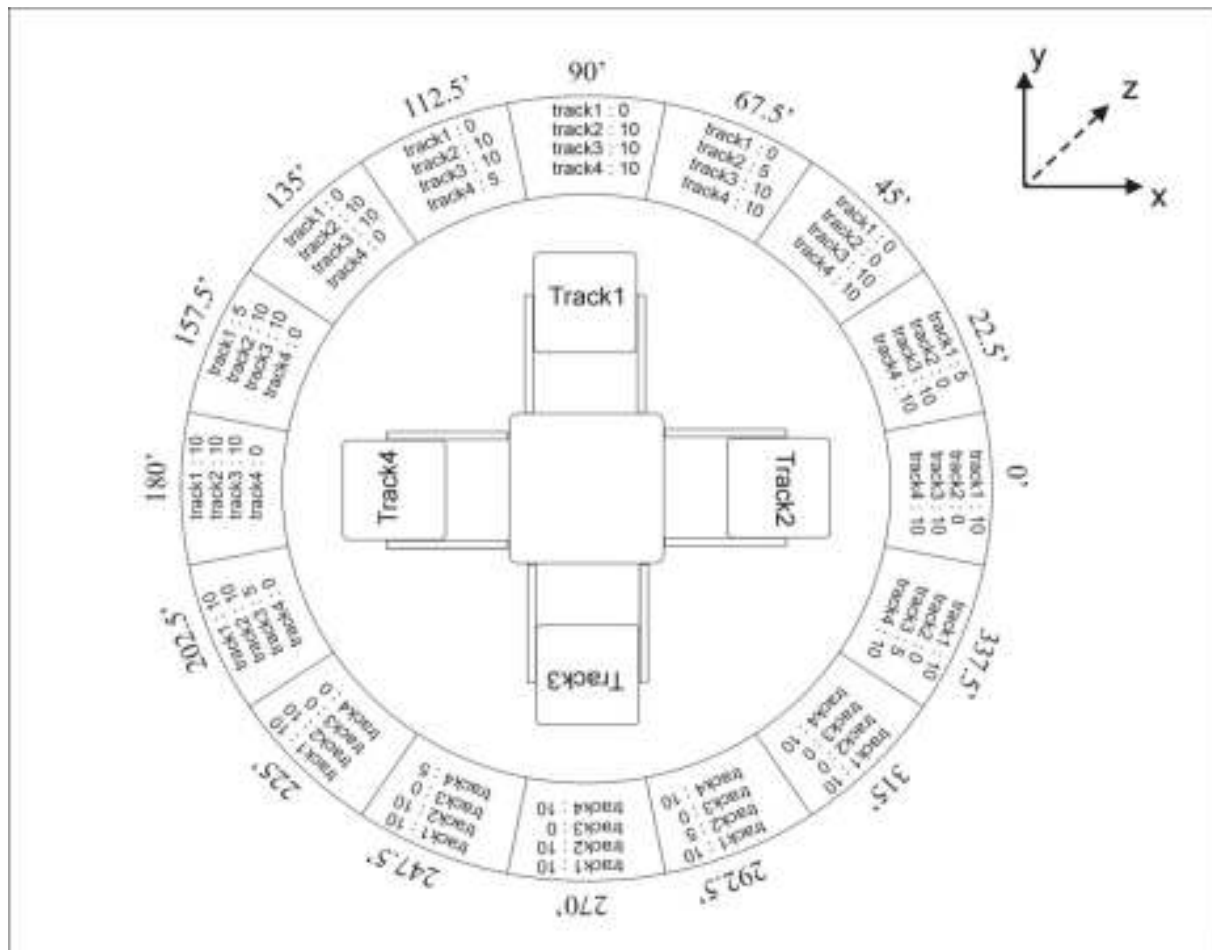


Figure 15. Mode of FAMPER's operation

In addition, the model allows FAMPER to adapt its control properties based on the environmental situations because control actions can be monitored and speed of each motor can

be changed by feedback results only. For example, when robot wants to turn 90 degree, caterpillar controller has to set that motor speed of track1 to 0, motor speed of tracks 2, 3 and 4 to 10 based on the property values described on the fig. 15 above.

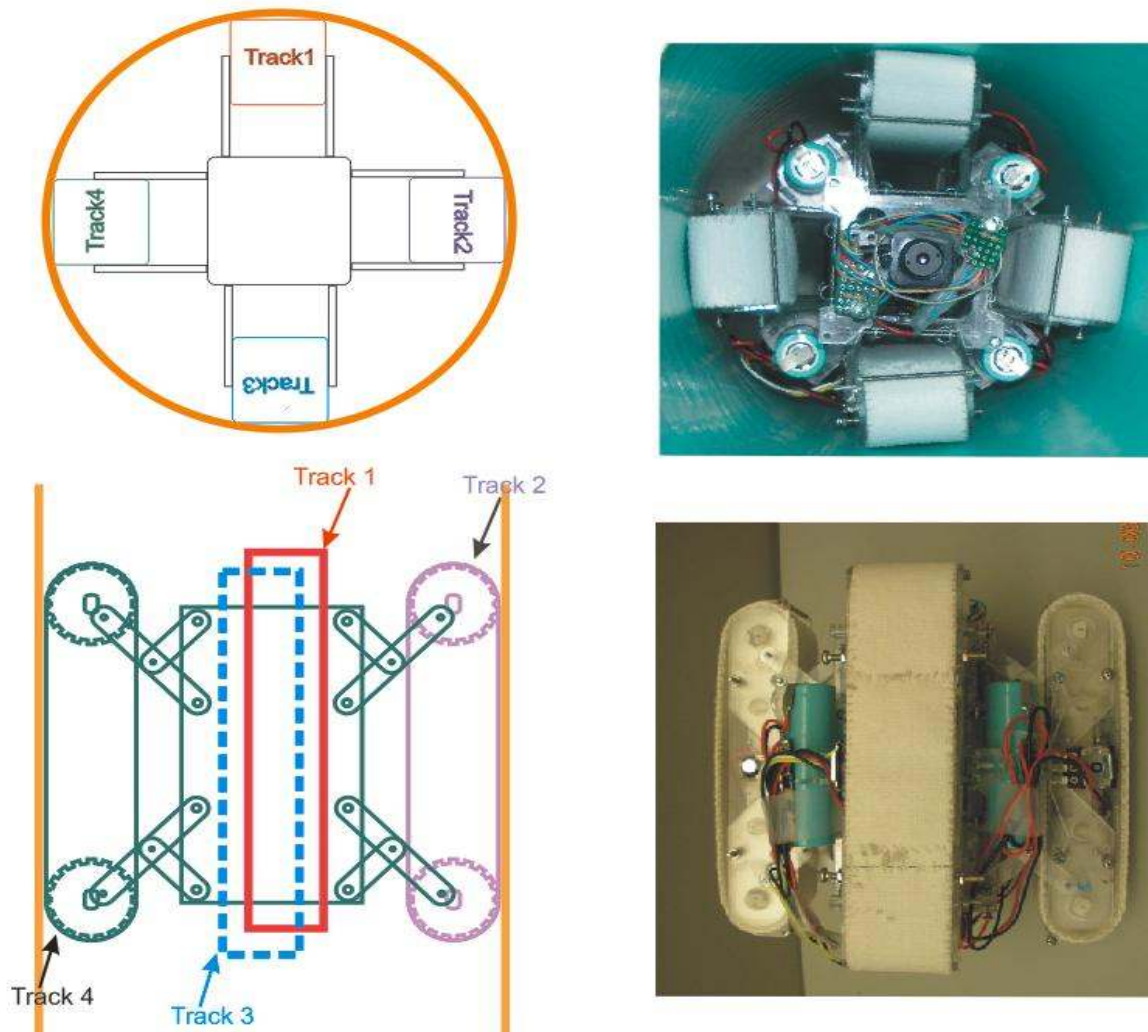


Figure 16. Front view of FAMPER

Let's say that Track 2 side of the inner pipeline-wall is contaminated by oil, and then the robot will be turning 45 degree instead of 90 degree because 50% of the rotation on track 2 slips, and speeds of tracks has to be changed. If robot keeps trying to turn with same combination of motor speeds, robot could be stuck in the pipeline and it could have been damaged. The proposed model can monitor this action and adjust the motor speeds based on the feedback that is

generated by Perception Operator. Below we describe the control mechanism in basic elbow type and branch type fittings of the pipeline.

Table 1. Forty five (45) degree elbow

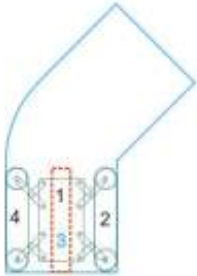
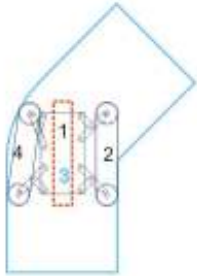
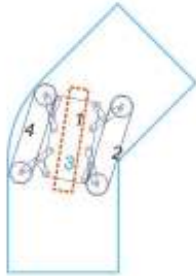
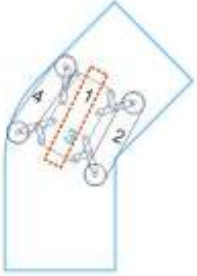
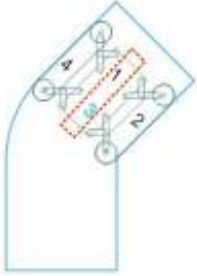
 1) All tracks turn at speed 10	 2) Track 2 reduce speed by 5 but the other tracks	 3) Track 2 reduce speed by 5 but the other speed 10
 4) Track 2 reduce speed by 5 but the other speed 10	 5) All tracks turn at speed 10	

Table 2. Ninety (90) degree elbow

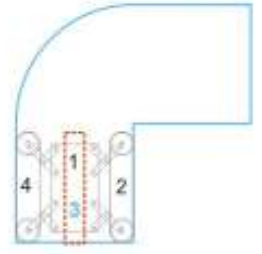
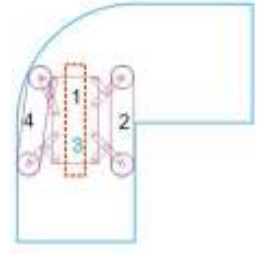
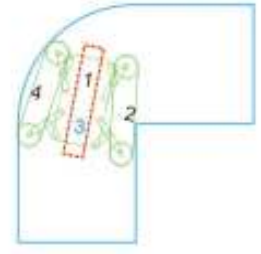
 1) All tracks turn at speed 10	 2) Track 2 reduces speed by 5 but the other tracks continue at speed 10	 3) Track 2 stops but the other tracks continue at speed 10
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Table 2. (Continued)

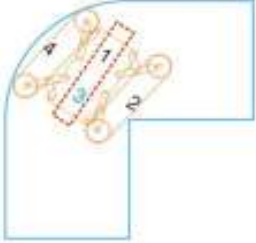
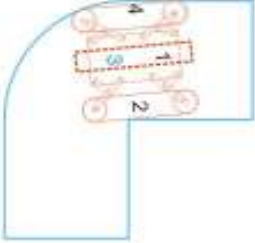
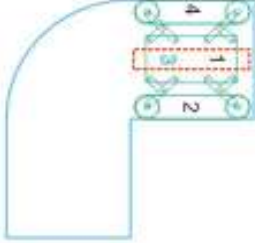
 4) Track 2 stops but the other tracks continue at speed 10	 5) Track 2 increases speed by 5 but the other tracks continue at speed 10	 6) All tracks turn at speed 10
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Table 3. T-Branch

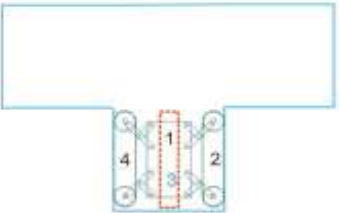
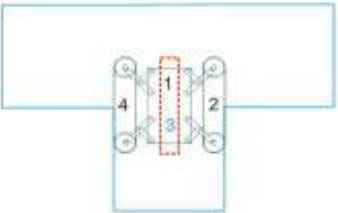
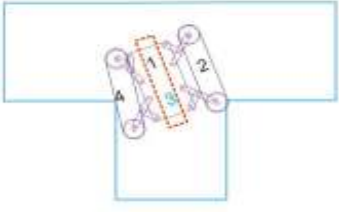
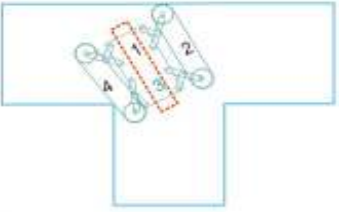
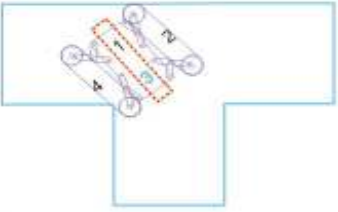
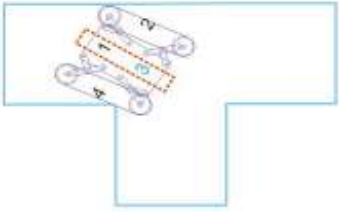
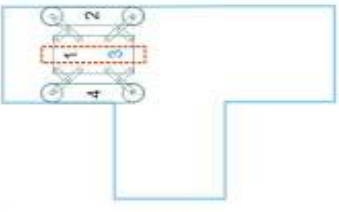
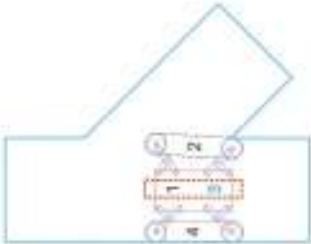
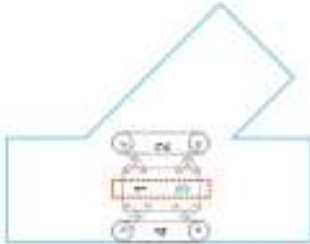
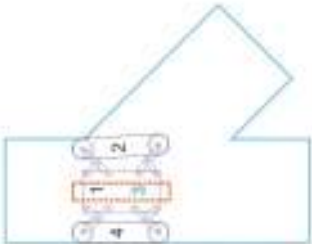
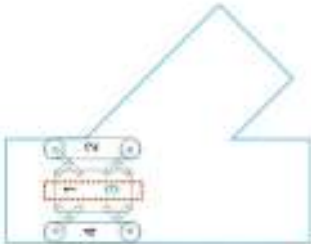
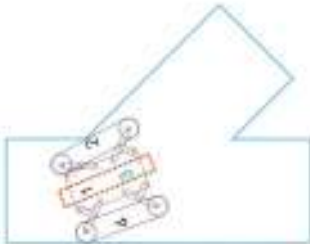
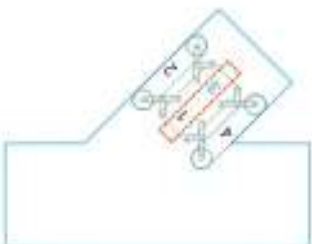
 1) All tracks turn at speed 10	 2) All tracks turn at speed 10	 3) Track 4 reduces speed by 5 but the other tracks continue at speed 10
 4) Track 4 continue at speed 5 but the other tracks continue at speed 10	 5) Track 4 stops but the other tracks continue at speed 10	 6) Track 4 make reverse turn at speed 5 but the other tracks continue at speed 10
 7) All tracks turn at speed 10		

Table 4. Y-Branch

 1) All tracks turn at speed 10	 2) All tracks turn at speed 10	 3) All tracks turn at speed 10
 4) All tracks turn at speed 10	 5) All tracks turn at speed 10	 6) Track 2 stops but the other tracks make reverse turn at speed 10
 7) Track 2 make reverse turn at speed 5 but the other tracks continue reverse turn at speed 10	 8) All tracks make reverse turn at speed 10	 9) All tracks make reverse turn at speed 10
 10) All tracks make reverse turn at speed 10		

3.5 FAMPER's Electrical Architecture

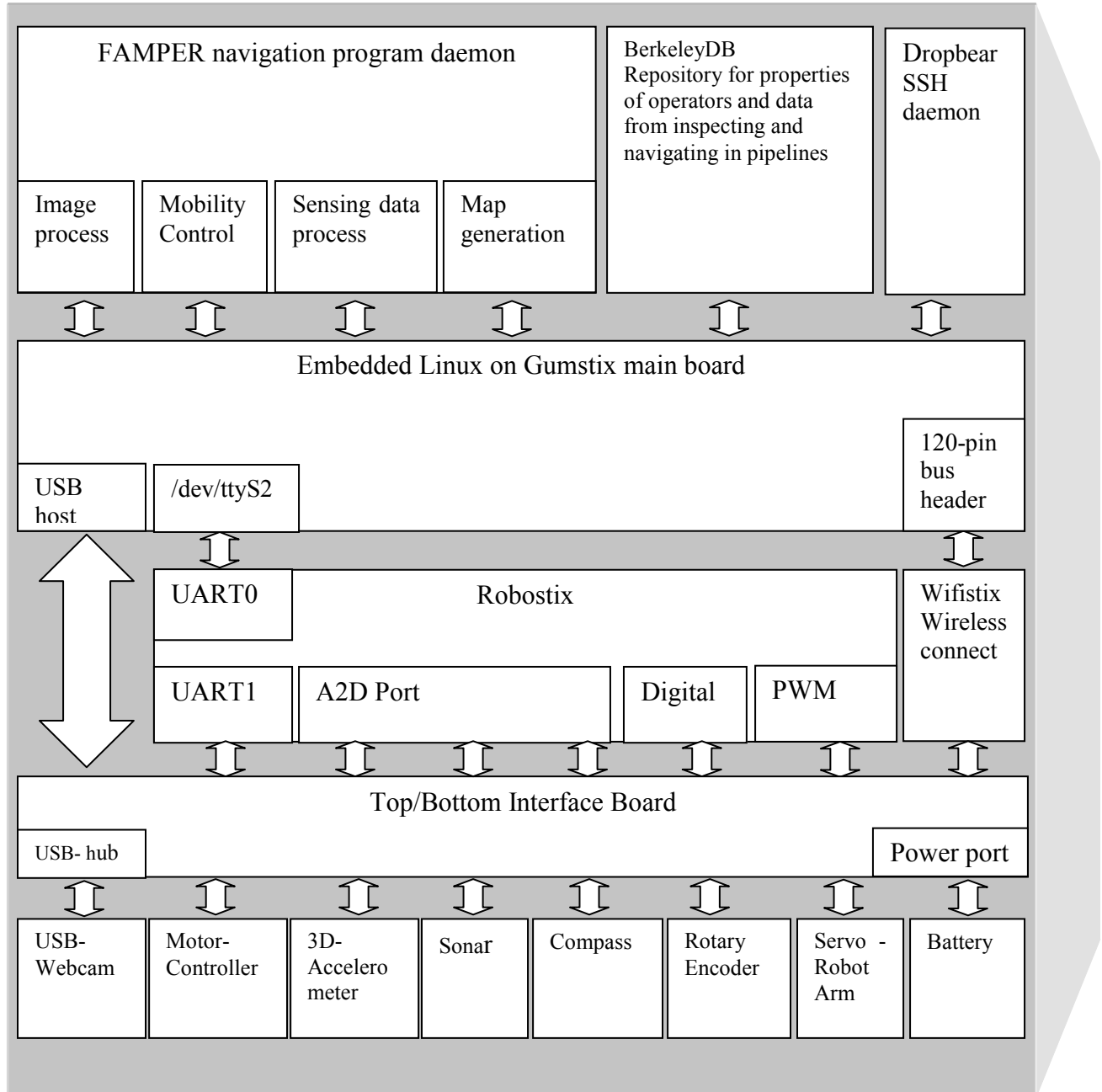


Figure 17. Block diagram of FAMPER's electrical architecture

3.5.1 Sensor and Controller

In the sensor and controller level of the electrical architecture, we will be implementing the sensors that will be used to set or modify the property values, which can be later used in

tuning the robot, so that the robot will make its movements according to the environment. We can use any type of sensor like sonar for scanning the surroundings. The sonar gives back the distance at which the obstacle or hole is, so we can save these values in the database and other operator can use this data for future actions. A compass can be used for obtaining the direction in which the robot is heading, a 3D accelerometer can be used to get the tilt information of the robot, and a rotary encoder can be used get the distance that the robot travelled thus far.

3.5.2 Interface Board

The interface board is used to provide the efficient and easy to reach connections to all the sensors and some controllers, so that even when one or more sensors are damaged then we can replace them without disassembling the entire robot. This interface board can also be used to connect all the expansion modules on to their respective interface board. And also it regulates proper voltages of all devices and removes noises from large current consuming from Motors.

3.5.3 Expansion Board

Although Gumstix is a small yet powerful motherboard, it does not include all the additional modules that we need for communicating with the gumstix, so in order to add these abilities to the gumstix, we can insert some additional expansion boards with the required capabilities. Some of the expansion boards, which we are using, are, WIFI-Stix, Robostix, GPS-Stix and AudioStix. Also these expansion boards can eliminate the limitations of gumstix to be used for robotics itself.

3.5.4 Gumstix Main Board

As already mentioned above, Gumstix is a small yet powerful motherboard which is flexible enough to make any kind of applications with it. The benefit of using gumstix is, because it is minute, we can use it in any small sized device and also as it is powerful enough,

the designed applications can take advantage of its processing power, by using high level languages, instead of programming in low level languages such as assembly language, PBASIC and etc.

3.5.5 Application

Using the benefit of gumstix architecture and Linux operating system, we can use high level language programs like Java, Python and etc. to make the applications secure, user friendly. Also with the use of open source operating system, we will always have the flexibility of adding and implementing any new applications on to the Linux platform. In FAMPER's architecture we will be using a central database to hold all the property values and sensor data from different operators.

With the proposed design, we can expect that FAMPER has excellent mobility in horizontal and vertical sections in a complex pipeline network, flexible crossing over obstacles, deformities or cracks that could happen in pipelines, and fully autonomous navigation system with its own processor.

3.6 Hardware System for a Fully Autonomous Navigating System

In order to satisfy with the proposed operation model, FAMPER's navigating system requires high performance processor and large memory in order to drive controlling robot with large sensed data. Especially, 6 inch pipe only allows very narrow choice in size of main board. In general, even the smallest sized main boards of desktop pc or laptop have more than 8 x 10 inch in dimension. Boards with these dimensions are too big to fit into 6 inch pipeline. Recently, mini sized motherboards have been available in the market. One of them is Gumstix which is a very small sized motherboard with powerful processor and big enough memory.

3.6.1 Gumstix

Gumstix is basically a ARM Architecture based motherboard, which can be used in every way we would use a regular computer with the similar potential. Gumstix based products are distributed as both cased packages, which comprises of the required peripherals for communication and also as a single board package, where we only have the motherboard. For a basic Gumstix board the I/O is usually provided using additional expansion boards. The gumstix motherboards started the evolution from Basix with a 4 MB flash, whereas the basix-xm models extend the flash to 16 MB, Connex motherboards also come with 16 MB flash whereas they have an additional bus header of 92 pins, which can be used for connecting additional expansion boards. The latest motherboard from gumstix is called Verdex and it comes in many configurations with variety of processor speeds and flash sizes. The key ability of a verdex board is its ability to be a USB host, higher RAM and higher flash memory.

Gumstix has created an extensive line of Xscale based processor and peripheral modules that can be used to create systems which are very compact yet very flexible. Robostix, which is also used in this project, is a module which when put together with the gumstix give us an enormous potential to build autonomous as well as controlled systems. The Robostix by itself provides connections for power and it has its own 8-bit AVR micro-controller.

Basically Gumstix is a single board motherboard which provides a lot of flexible options for Robotics or any other field you could think of. These computers are pre-installed with a version of the Linux 2.6 (Kernel) Operating System and have a variety of feature-rich expansion components for adding WiFi, BlueTooth, Ethernet (RJ-45), and many other capabilities. Though small, speeds of these computers range between 200 and 600 MHz and support 64 or 128 MB of SDRAM. They do not support hard disk drives but store data in Flash memory available from 4

to 32 MBs. Though the Gumstix come preloaded with Linux, Windows CE has been successfully placed on Gumstix. The various configurations of CPU, RAM, and Flash memory storage are based on the design of three different types of motherboards. Because of the difference between motherboard designs, many expansion components are compatible with only certain models.

Table 5. Benchmark result of Gumstix PXA-270 with Pentium 90 and AMD K6/ 233

Operation Performed	Iterations /sec.	Pentium 90 / Gumstix 600HMz PXA270	AMD K6-233 / Gumstix 600HMz PXA270
NUMERIC SORT	198.12	5.08 / 1	1.67 / 1
STRING SORT	17.469	7.81 / 1	1.21 / 1
BITFIELD	6.6802e+07	11.46 / 1	2.39 / 1
FP EMULATION	41.583	19.95 / 1	4.60 / 1
ASSIGNMENT	1.2438	4.73 / 1	1.23 / 1
IDEA	538.49	8.24 / 1	2.45 / 1
HUFFMAN	302.12	8.38 / 1	2.68 / 1
Baseline: Pentium 90 with 256 KB cache running MSDOS			
Baseline: AMD K6/233*, 512 KB L2-cache, gcc 2.7.2.3, libc-5.4.38, Linux 2.0.32			

As we can see in table 1, 600 MHz PXA270-based gumstix is generally twice as fast as 233 Mhz K6 or about 8 -10 times the speed of a Pentium 90. [104]

4. IMPLEMENTATION OF FAMPER

This particular implementation of FAMPER focuses on verification of mobility in vertical and horizontal pipelines with manual controlling system based on FAMPER's operational architecture using Gumstix based devices. In the FAMPER's operational architecture, I completed implementation of action operator without Supreme and Perception operators. Other operators can be developed independently with minor changes in the action operator, which I already implemented. This wouldn't have been possible if FAMPER was not using modular architecture.



Figure 18. Initial Structural Design of FAMPER



Figure 19. First prototype of FAMPER



Figure 20. FAMPER version 1.0

4.1 FAMPER's Disassembly

FAMPER is designed as a competent fully autonomous and un-tethered robot which can navigate through 6 inch sewer pipelines in both vertical as well as horizontal situations involving all types of special fittings for a given pipeline network. The fore mentioned FAMPER's capability for autonomy and its mobility can remarkably extend its potential to larger mission ranges, when comparing with other traditional inspection equipment that is limited by the cable and poor mobility. FAMPER is equipped with all the required resources in its body, like sensors, batteries, video camera, and main control board. FAMPER is ready with Four Li-ion 2400mA batteries that supply power to 8 motors, a RF camera, two LED-lights, and a Gumstix based

computer system which includes wifi-connection module and sensors, supporting about an hour for its mission performance.



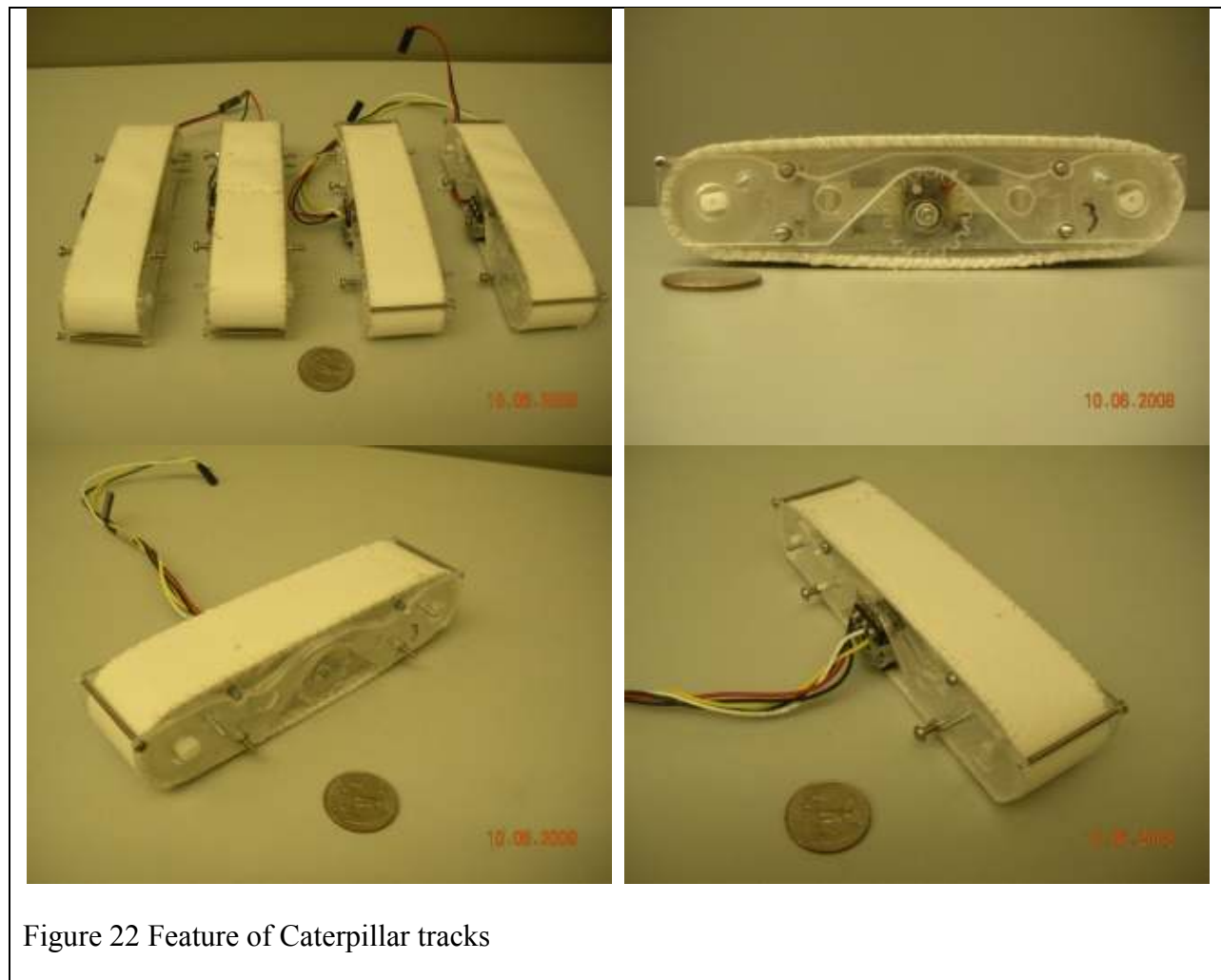
Figure 21. Disassembly of FAMPER

4.2 FAMPER's Mechanical Platform

All commercial sewer pipe robots are capable of moving in straight pipes but not in any kinds of bended or branched pipes. Moreover, most of those robots are not capable to travel in vertical pipeline network. However, mobility in vertical pipelines is one of biggest challenge for a sewer pipeline robot, because most of sewer pipelines in general houses and buildings consist of very complicated network, including vertical as well as horizontal pipelines. In the mechanical model of FAMPER, wall-press type is considered for vertical mobility and caterpillar type is

combined with the former for increasing the gripping force.

4.2.1 Caterpillar



As we know that a caterpillar-based wall-press robot would be much more efficient for a sewer pipeline system, I have implemented 4 caterpillar tracks each 33mm wide and 148mm in length. Each caterpillar track is equipped with a rotary encoder to calculate the distance moved thus far and also as an added advantage we will be aware, if one or any of the tracks are not moving as they are expected to be. So these tracks can be tuned for future operations. For giving better torque for the rotation, each caterpillar track has 2 motors attached one on each side of the track.

4.2.2 Flexible and Independent Links and Suspensions

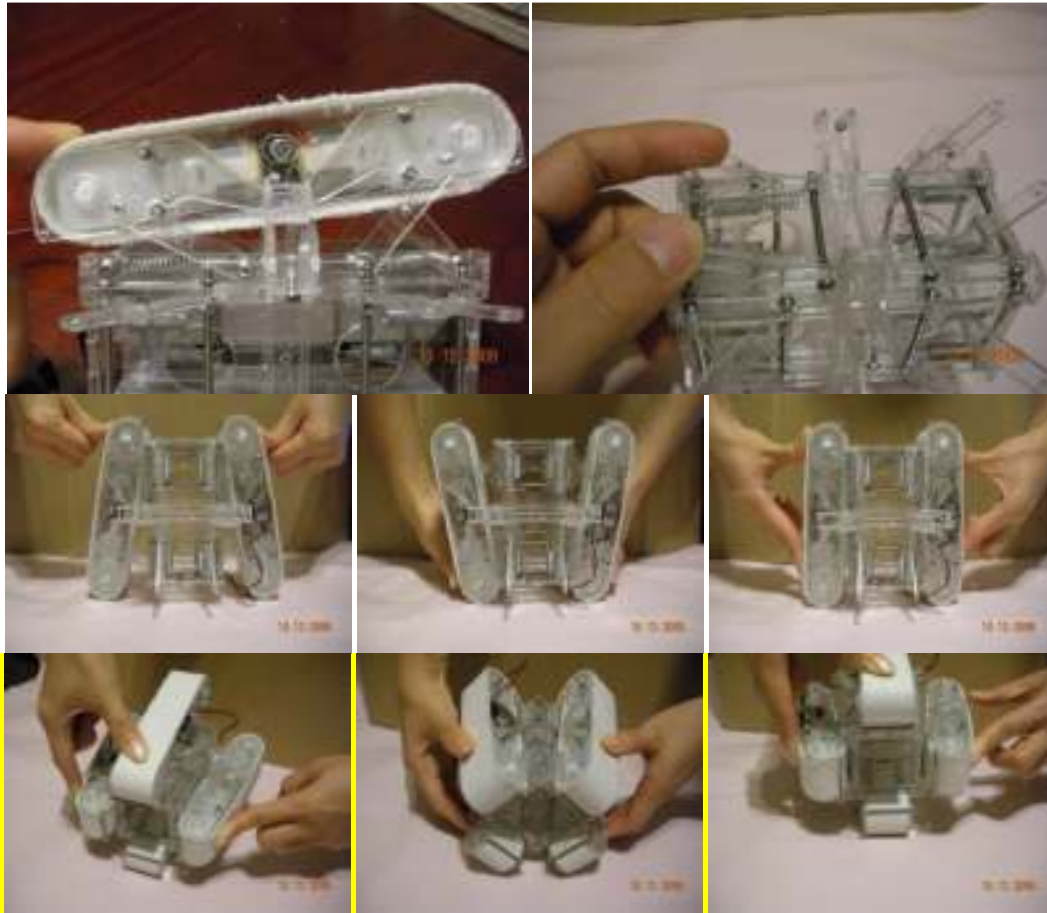


Figure 23. Feature of links and suspensions

Along with the caterpillar system, we also need the wall-pressing ability for the robot to be able to climb up in the vertical situations as well. Each caterpillar track is mounted to the central body using 4 independent suspension links. These links are responsible in giving the required gripping force to the robot and also the robot can be contracted from 157mm to 127mm using these links. This ability of being able to contract and expand will make the FAMPER in being flexibly moving through disfigured pipelines as well.

4.2.3 Skeletal Body Frame

As we discussed previously that spacious body can give us the capability of using better computing devices and sensors, so the central body frame of FAMPER has 40mm X 40mm X

108mm spacious room, which is enough to fit gumstix, robostix, 2 interface boards, wifi-stix and the additional sensors that we have used in this implementation of FAMPER. Also the front and rear vents of the central body gives us easy access to all the ports of the interface boards.

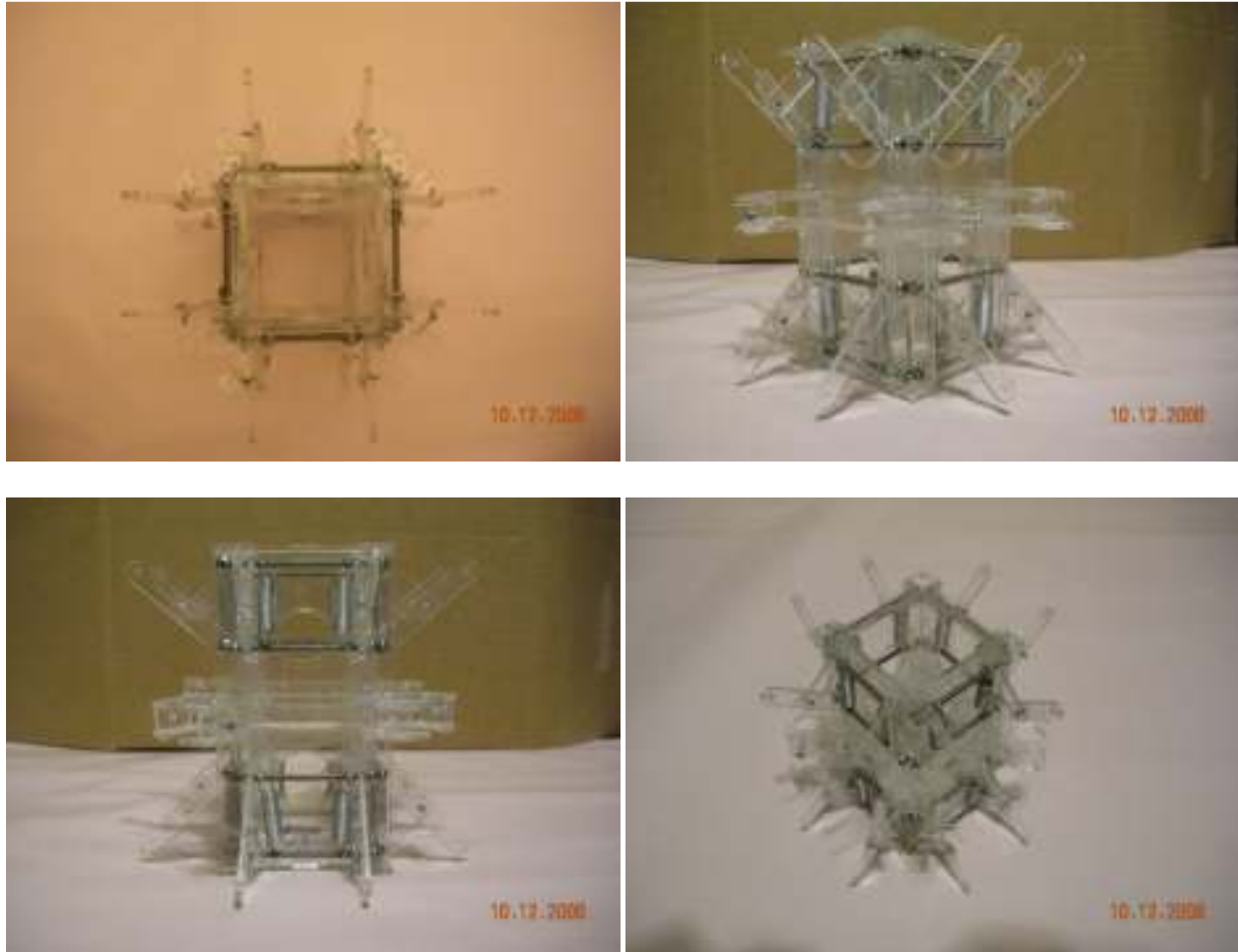


Figure 24 Feature of central body frame

4.3 FAMPER's Electrical Platform

FAMPER's small and powerful computing system is sandwiched between the two interface boards, which are in turn connected to all the internal ports of the gumstix and the robostix. Also the power for all the internal boards will be regulated by the bottom and top interface boards. This layout gives us easy and universal access to all the ports that we need at easily accessible places like, front and rear of the central body. The computing system consists of

a stack of 5 boards, which are bottom interface board, Robostix, Gumstix, Wifi-stix and top interface board from bottom to top respectively.

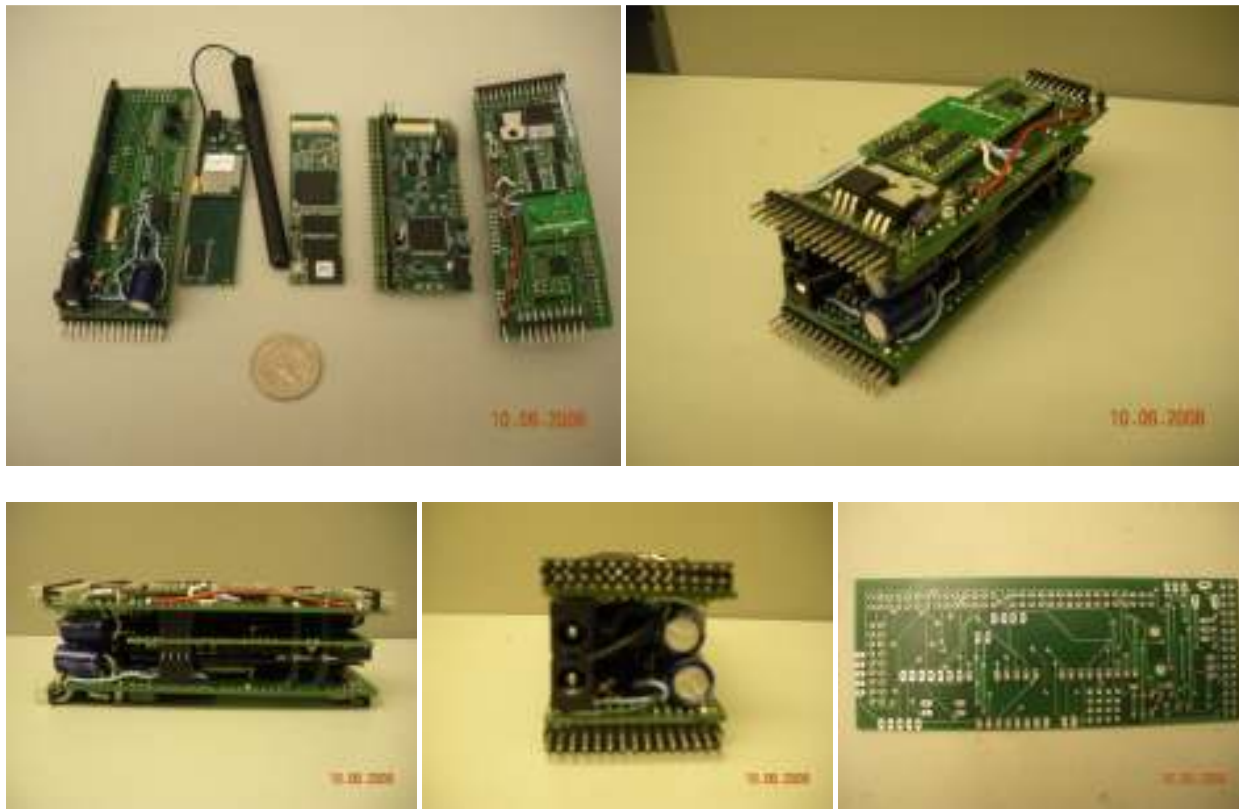


Figure 25 FAMPER's electrical disassembly

4.4 Electronic Architecture

FAMPER has been implemented with a RF-CCD Camera which can send the video stream independently without any intervention from other modules implemented. The sensors that are included in this particular implementation of FAMPER are 3D-Accelerometer, Compass and Rotary Encoder. Here, 3D accelerometer will give us the tilting information of the robot, compass gives us the direction on which the robot is heading and rotary encoder will calculate the distance travelled by the robot so far.

The interface board allows convenient connectivity of these sensors to the main board and supply the required power to those modules. The Gumstix based expansion modules used in

this implementation are Robostix, WIFI-Stix. Robostix is a module that will provide the required pins to read data from these sensors, by converting them from analog to digital and then forwarding them to the gumstix. Now the Gumstix has the module called WIFI-Stix which will add the functionality of transmitting and receiving all of this data wirelessly on to another remote computer.

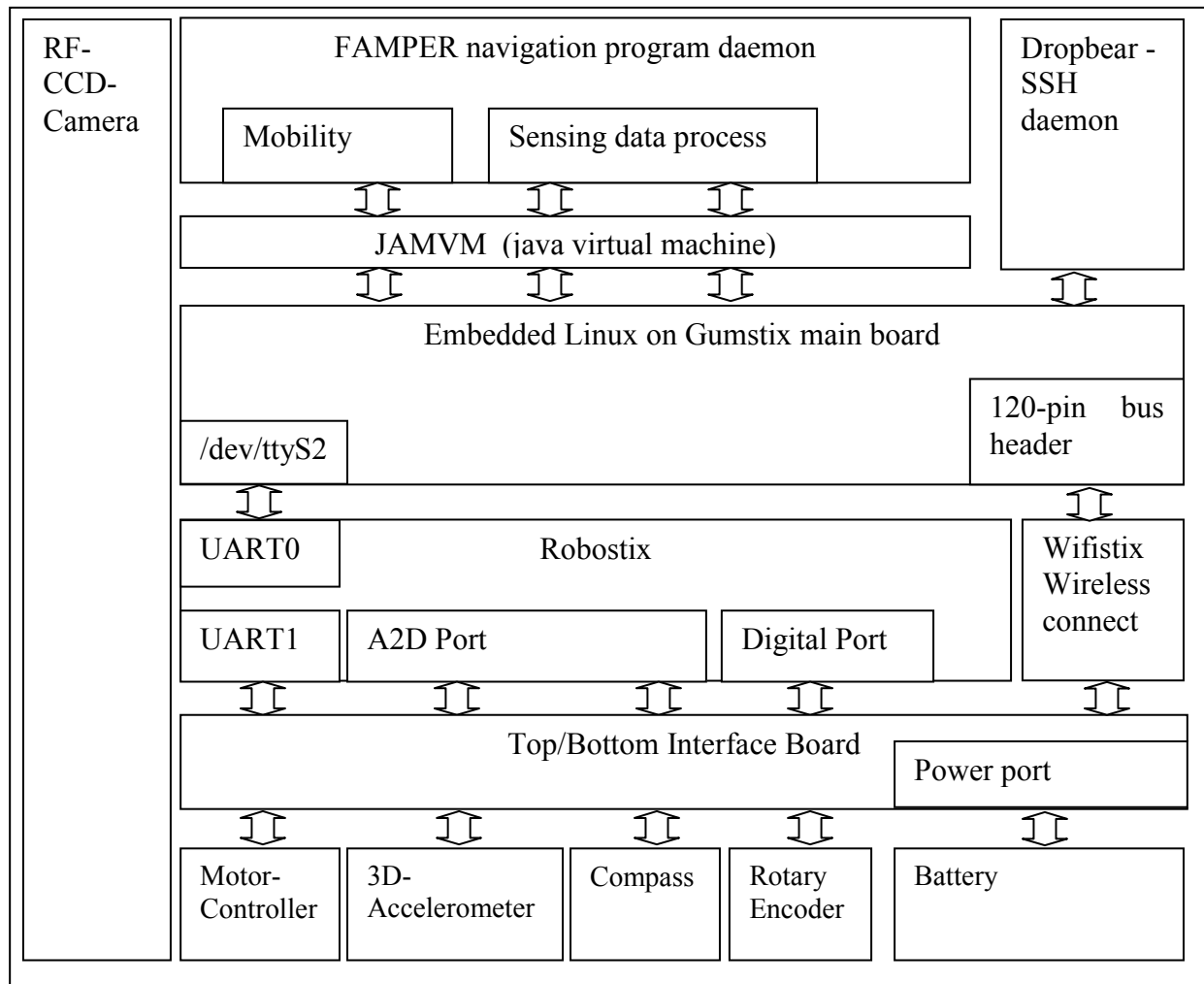


Figure 26. Block-Diagram of implemented electrical architecture

Gumstix is the main board which will host the applications that are supposed to be running for the successful execution of other modules. Gumstix is installed with Buildroot based linux platform which has the capability of running programs written on high level languages like Java, Python and Perl. In this implementation I programmed the controlling interface using Java.

The sensor readings have been read using programs written in AVR-C, which are then ported to the robostix.

4.5 FAMPER Manual Control Program

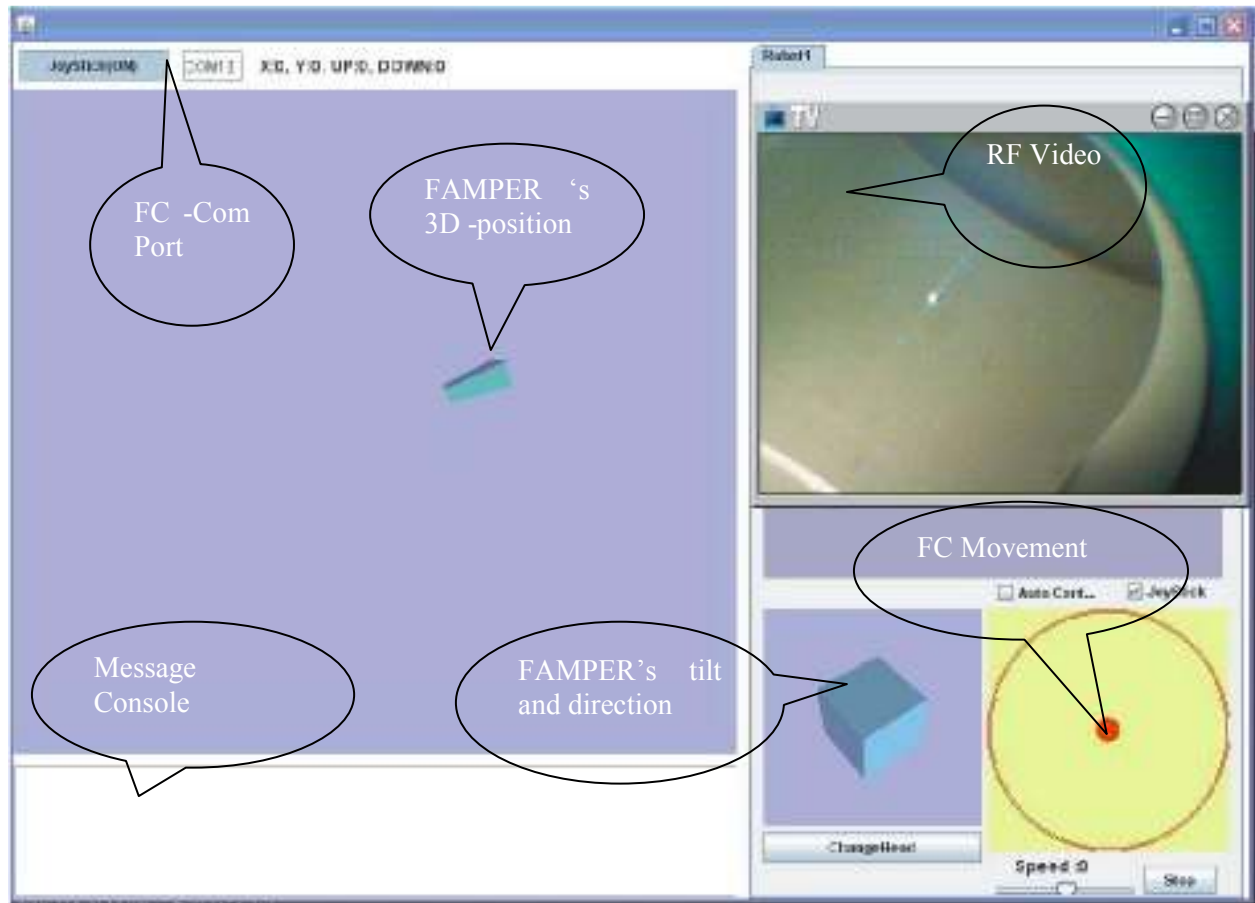


Figure 27. Feature of manual control program

In this implementation I have built a manual control interface program, which can aid us in operating the FAMPER in most of the pipeline layouts. The interface mainly comprises of four major panels, 3D view of FAMPER's positions, RF video panel, which can display video stream from the FAMPER's RF camera, Control Panel, which gives us the ability to control the FAMPER by using FC or even by using the GUI interface and also the tilt and direction of FAMPER are indicated in mini 3D view, and Message Console which can display the detailed information of the status of FAMPER.

4.6 FAMPER Controller: FC

The controlling signals for the robot can be passed to the computer running the FAMPER's Manual Control Program using a FAMPER Controller (FC). FC uses an analog joystick along with flexibility of sending the control signals using Bluetooth or USB connections. It has been programmed using C for PIC (Programmable Intelligent Computer) microcontrollers, PIC16LF88.



Figure 28. Simple FAMPER Controller: FC

4.7 RF Video System

In order to control the FAMPER from a remote location we need a visual system which can give us the ability to see and control the FAMPER. In this implementation I used RF Video system, which is small in size, low in power consumption and has high sensitivity feature that is required for us to imagine the situation inside the pipeline. I have also added ultra bright LED's which can give the RF Video camera system to capture the video. This RF video camera is completely independent from the FAMPER's main electrical system.



Figure 29. RF video system

5. EXPERIMENTAL EVALUATION

5.1 Elbow – 45

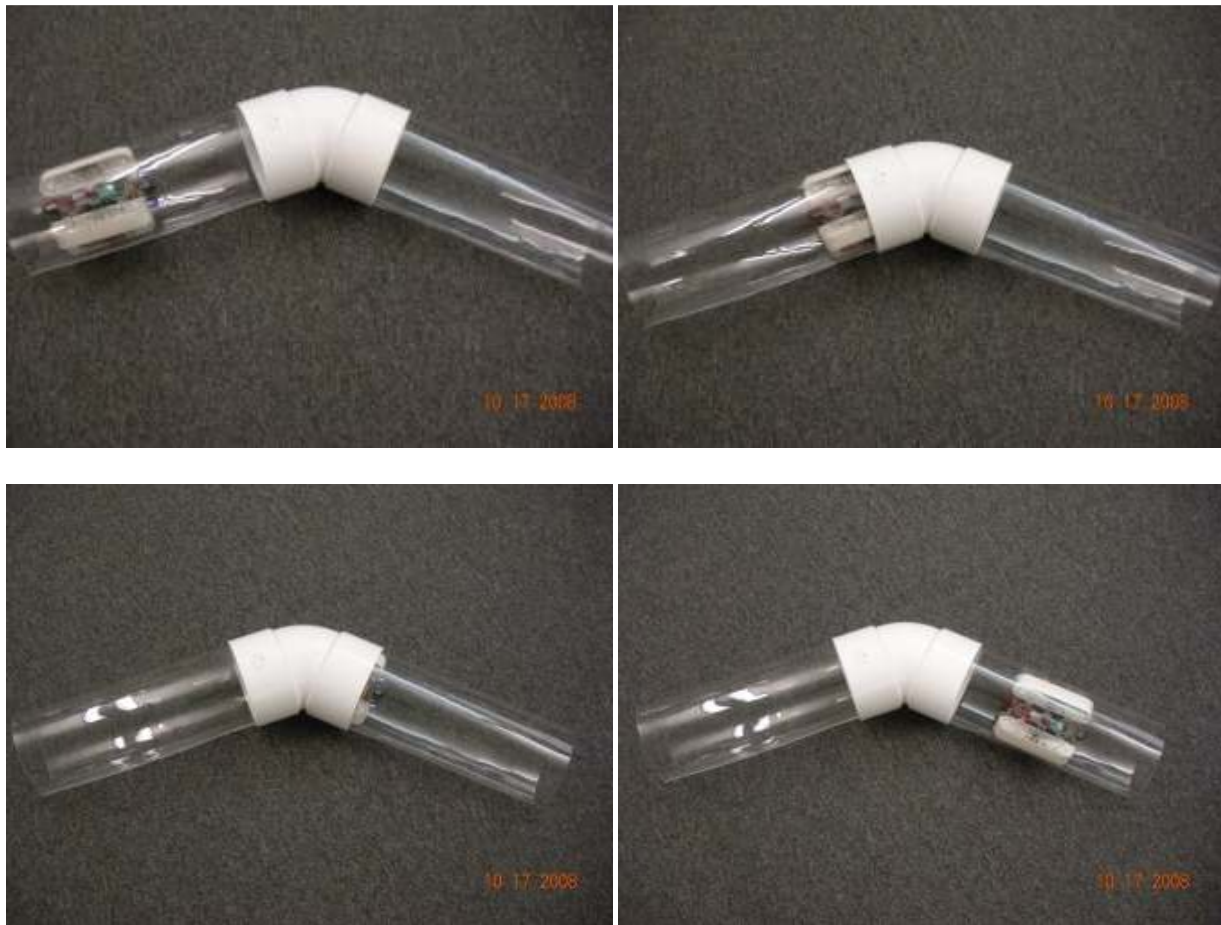


Figure 30. Test-bed of 45 degree elbow

As we can see from the fig.30 above, the robot starts travelling in straight pipeline and makes a transition to a 45' and back to the straight pipeline. Now by changing the direction of the motors we can go backwards by following the exact path.

5.2 Elbow – 90

As we can see from the fig. 31 above, the robot starts travelling in straight pipeline and makes a transition to a 90' and back to the straight pipeline. Now by changing the direction of the motors we can go backwards by following the exact path.



Figure 31. Test-bed of 90 degree elbow

5.3 Branch T

As we can see from the fig. 32 above, the robot starts travelling in straight pipeline and makes a transition to right of a T branch, in the similar fashion as it would for a 90' turn and back to the straight pipeline. Now by changing the direction of the motors we can go backwards by following the exact path until the T branch, and then following the straight path ahead.

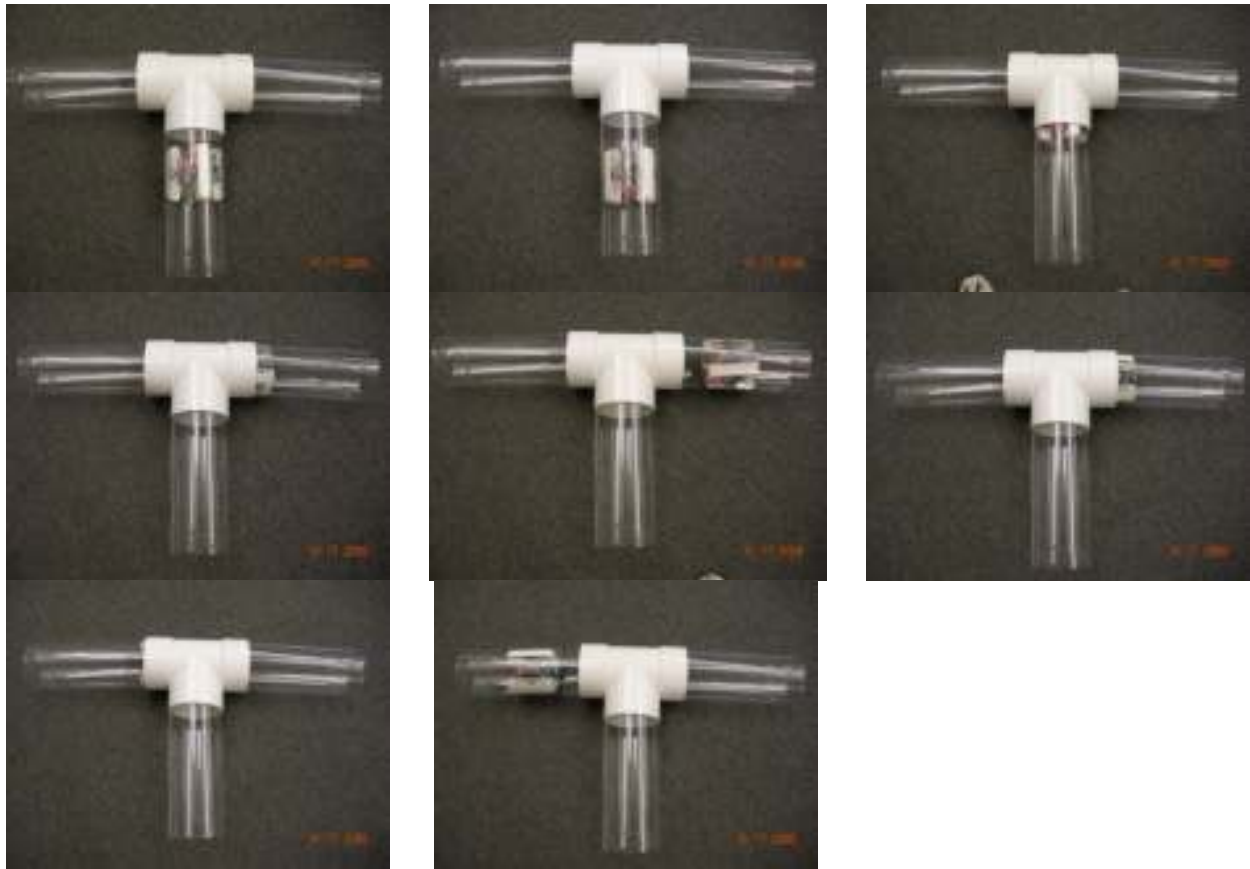


Figure 32. Test-bed of branch-T

5.4 Branch Y

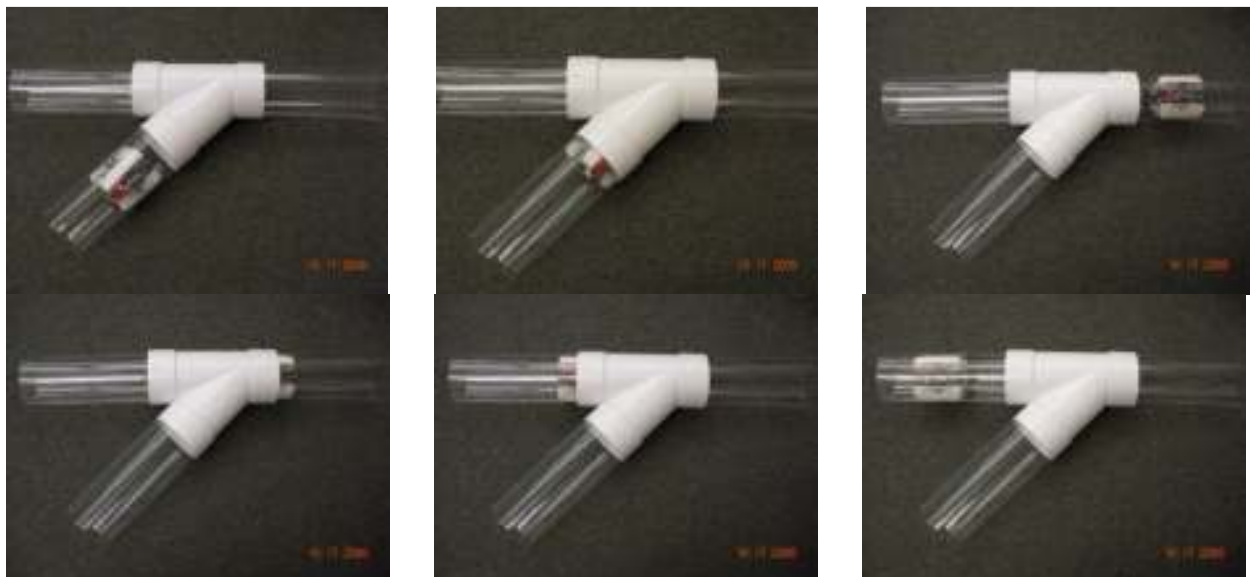


Figure 33. Test-bed of branch-Y

As we can see from the fig. 33 above, the robot starts travelling in straight pipeline by taking the path to the right of the Y-branch, and on its way back to the Y branch it moves ahead on the straight path to meet the straight pipeline on the other end.

5.5 Complex Pipeline Layout

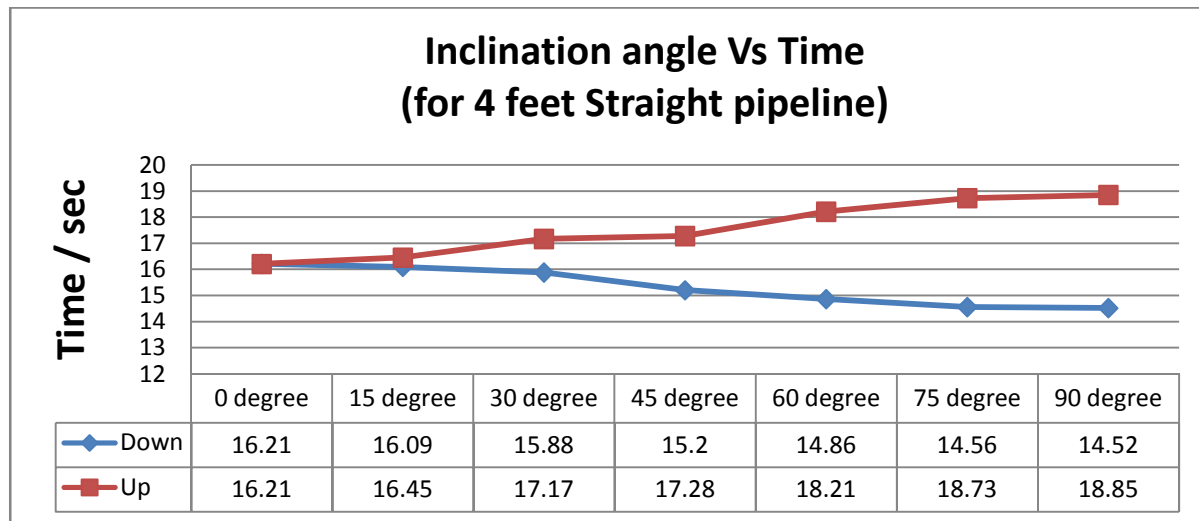


Figure 34. Test-bed Complex pipeline layout

The pipeline layout has been constructed using the available special fittings that a typical pipeline layout uses. FAMPER can travel through the complete pipeline layout by changing the motor speeds as described in Chapter 4.

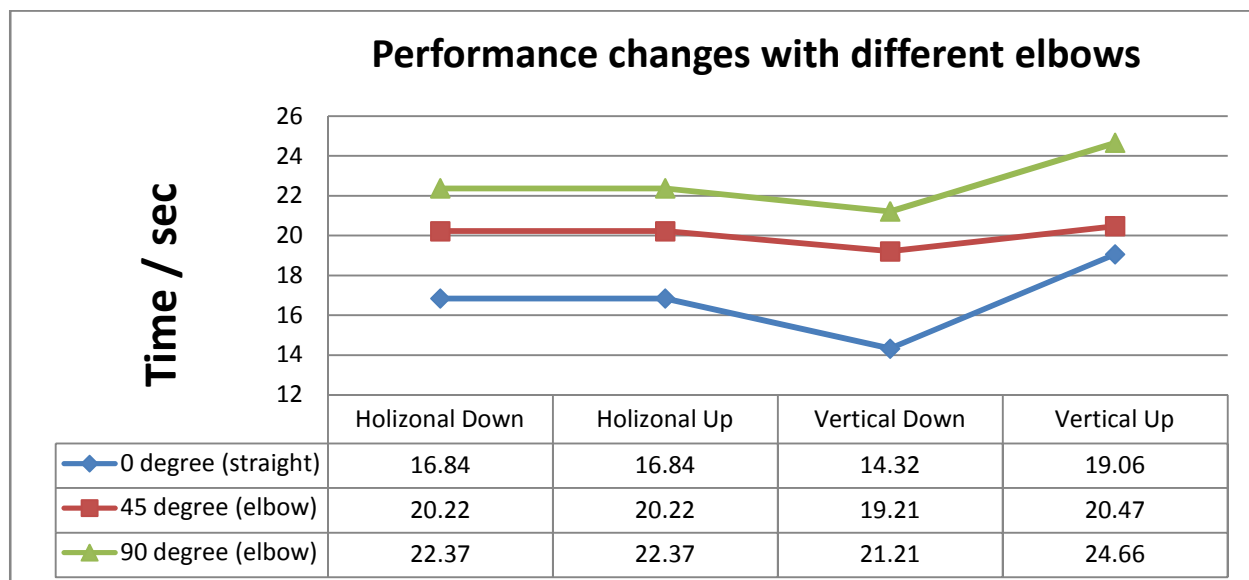
5.6 FAMPER's Performance Result

Table 6. Inclination angle Vs. time for 4 feet straight pipeline



I have evaluated the FAMPER's performance in crossing a straight pipeline from one end to the other at different inclination angles, to see how the speed would be affected with the inclination angles. As we can see from Graph 1 quite apparently, as the inclination angle increases from 0' to 90', the time taken by FAMPER to reach the other end for a downhill case, decreases and for an uphill case, increases.

Table 7. Performance change with different elbows in 4 feet length



I have compared the FAMPER's performance in crossing different types of elbows from one end to the other through the elbow. All the pipelines considered for this experiment are of the same length. As we can observe from the Graph 2, the time taken for completing the trip from one end to the other increases with the bending angle of the elbow, because, in elbows with different bending angles FAMPER has to adjust the speed of motors in different ways as mentioned earlier in Chapter 4, in order to successfully pass the elbow. And also we can observe that the horizontal and vertical conditions behave in the similar way as it did for the previous experiment.

We can conclude from the experimental results that FAMPER can effectively navigate through most of the pipeline layouts involving both horizontal and vertical pipelines. Even when the gravitational force, frictional force hinders the vertical movement of a robot, FAMPER was able to successfully overcome these with minor performance changes.

6. DISCUSSION

Although FAMPER has been designed and developed for specific purpose, the concepts discussed in this thesis can be extended for many other applications which require these functionalities. And in this implementation of FAMPER, we have proved that it did accomplish the fundamental requirements for transforming it into a fully autonomous robot, i.e. vertical and horizontal mobility, flexible control, powerful computing system, sensitive sensors. Also we have encountered some issues that we would like to discuss about.

6.1 Mechanics

Many robots have been designed so far in order to achieve their specific tasks. But most of these robots are designed with a limited perspective. In other words, a robot designed by a mechanical engineer might not consider or give much weight to the electrical and software issues and vice versa. FAMPER has been designed in such that it could take care of the limitations imposed for those robots. So, these issues can impose additional limitations in the design phase, like we need to allocate enough room for the electrical compartment and also for the mechanical structure, without neglecting the overall size specifications. We have encountered one such problem by miscalculating the length of the caterpillar links by 2 mm which in turn posed problem navigating through Y-branches. So we can have an idea of how a difference in millimeters can cause malfunctions in miniature sized robots. We have to decide if we need more flexibility or more functional performance by adding Artificial intelligence to the robot, this can be a debatable issue on whether to sacrifice flexibility or the functionality. As we are talking about a fully autonomous robot, sacrificing the functionality could make it a failure even before starting it, so here in the design of FAMPER we have sacrificed a little bit of flexibility to make it a strong contender to be a fully autonomous robot, although we do not need all the

computational prowess that FAMPER offers in this implementation for basic navigation of a robot.

6.2 Electronics

Power is one of the main issues which could limit the ability to use the robots in many applications. FAMPER is designed to run on 4 batteries, which are the only source of power for the entire robot, both the mechanical and electrical devices. Most of the power from the batteries is used for the operation of the motors and the rest for the computational devices. So, one of the core challenges that we should be taking care of in the future implementations is to make FAMPER intelligent enough to avoid taking the paths which are already explored, so in a way we will be saving the power for exploring the unexplored paths of the pipeline. From the experiments conducted using the present implementation of FAMPER, we have observed that motors consume about 75-80 % of the power from the batteries, so we can be aware of how important it is to regulate this to an efficient level.

6.3 FAMPER System Programming

As we mentioned previously we need a good Artificial intelligence system to make the robot both efficient in performance and the overall power consumption. So far many different algorithms have been proposed in normal field operations such as Unmanaged Autonomous Vehicle, Unmanned Aerial Vehicle, etc. These algorithms can be extended to the pipeline situations, but might not be as efficient as they were for the designed purposes. This leaves us an opportunity to develop new artificial intelligence algorithms for pipeline structures. We have also considered of implementing multiple robots, so that they can do the given job in co-ordination with each other robots. This will help us save the time taken to complete the given mission. The multi-robot strategy can also be used in rescuing other robots in certain conditions

which we will be discussing shortly. Also we need to add the required functionality in order for the robot to get back to the dock or a predetermined location in case of low power conditions, which will basically help it in not creating problems while trying to solve them.

6.4 Environmental Concerns

When working with real world pipeline structures, the inner pipeline conditions could be uncertain to the robot, such as residual water in the pipelines, disfigured pipelines, etc. Especially water could cause unrecoverable damage to the robot if it is not protected from the same. So we have to consider this issue by making electrical compartment waterproof. FAMPER has been designed to have a homogenous electrical body that could be enclosed into a waterproof capsule with minor changes. Also malfunctioned or damaged robot could hinder the operation of the pipeline by obstructing the path; we can clear this path with the help of other autonomous robots similar to the way the rescue team save the subject in trouble.

So in order satisfy all these limitations we need to come up with a truly efficient algorithms to take care of the same. Thus we can confidently propose that these could provide the required springboard to FAMPER, in order to take its first plunge as a fully autonomously mobile robot.

7. CONCLUSION

Robots have been used in many areas to replace the human efforts. Specially, pipeline maintenance is challenging area for which robots can be employed. Many robots have been developed for various pipeline based applications. There are sound ways to inspect pipelines; In-pipe inspection is one of them, on which I have focused my research. This thesis pays attention to the concept and design of mobile autonomous robot to explore pipelines by browsing them. A fully autonomous mobile pipeline inspection robot should have five principle features; firstly good mobility in the pipeline network is a must, secondly good sensing ability for pipeline inspection, thirdly good artificial intelligence for autonomous navigation, fourthly powerful computing system for processing large sensed data and sophisticated control and lastly powerful battery to increase its potential mission range. In this thesis, I propose design of a fully autonomous mobile pipeline inspection robot, named FAMPER. I have also described the mechanism that FAMPER provides for excellent mobility in vertical as well as horizontal pipelines, and proposed the system architecture that would enable FAMPER to be fully autonomous. In this implementation, FAMPER is equipped with a small yet powerful computing system, which allows it to be capable enough to execute complicated artificial intelligence algorithms. In addition, it has easy and extendable interfaces for various sensors. In the experiments, FAMPER showed excellent mobility in 6 inch sewer pipeline layout using its wide caterpillar tracks, independent flexible links and thanks to the sophisticated control mechanism.

During evaluation, certain limitations in control of FAMPER while navigating through T and Y branches were found. But it is a minor implementation problem, which can be easily remedied by making simple modifications to the caterpillar track. The main challenge that we could be facing is to completely implement the mentioned operational architecture of FAMPER,

to make it fully autonomous robot. Also we have to consider the pipeline inspection methodology which can increase the FAMPER's application areas.

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APPENDIX I : SCHEMATICS OF FAMPER

A. Mechanical Schematics

A-1. Robot Body

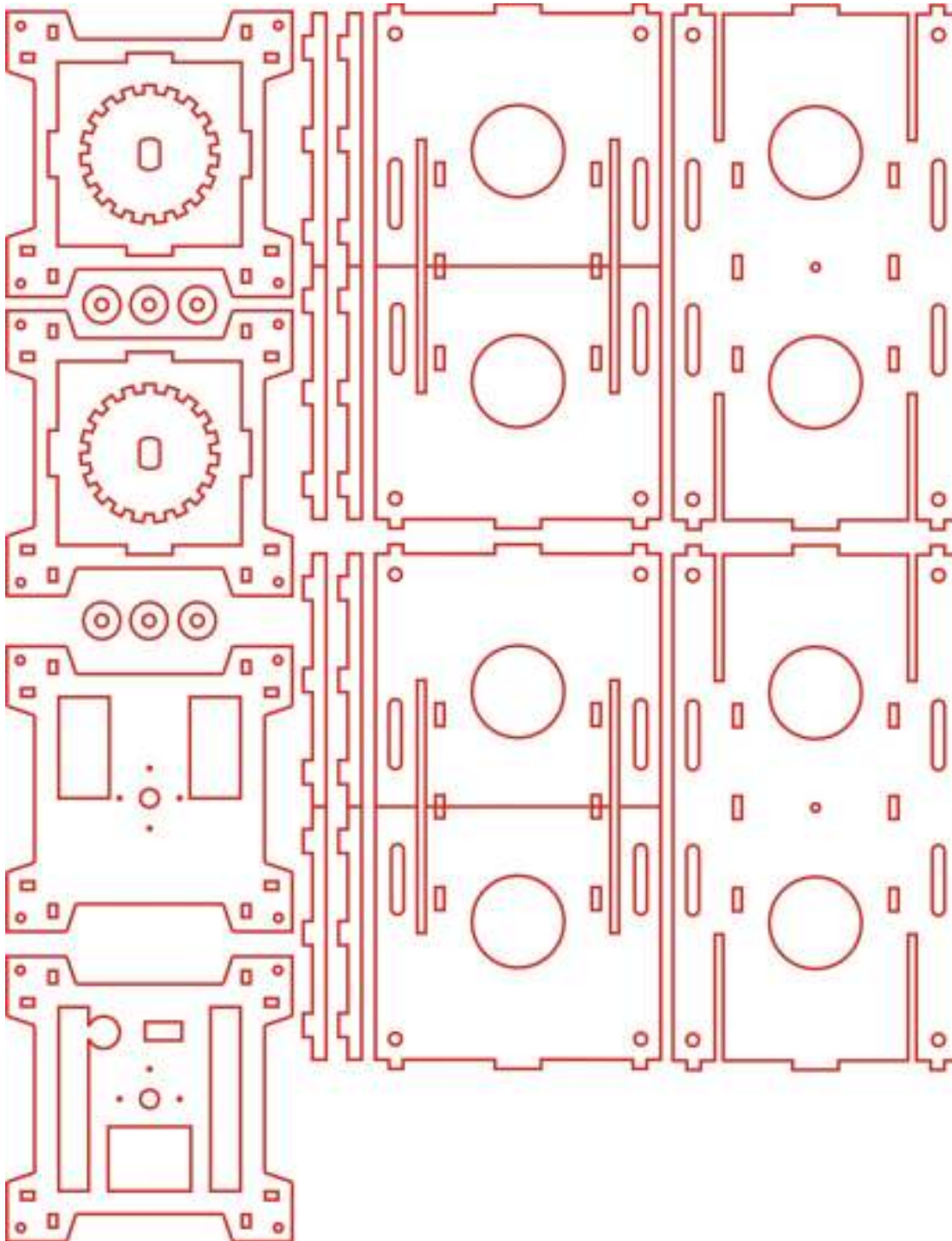


Figure A.1- Robot body mechanical schematics

A-2. Caterpillar Frame

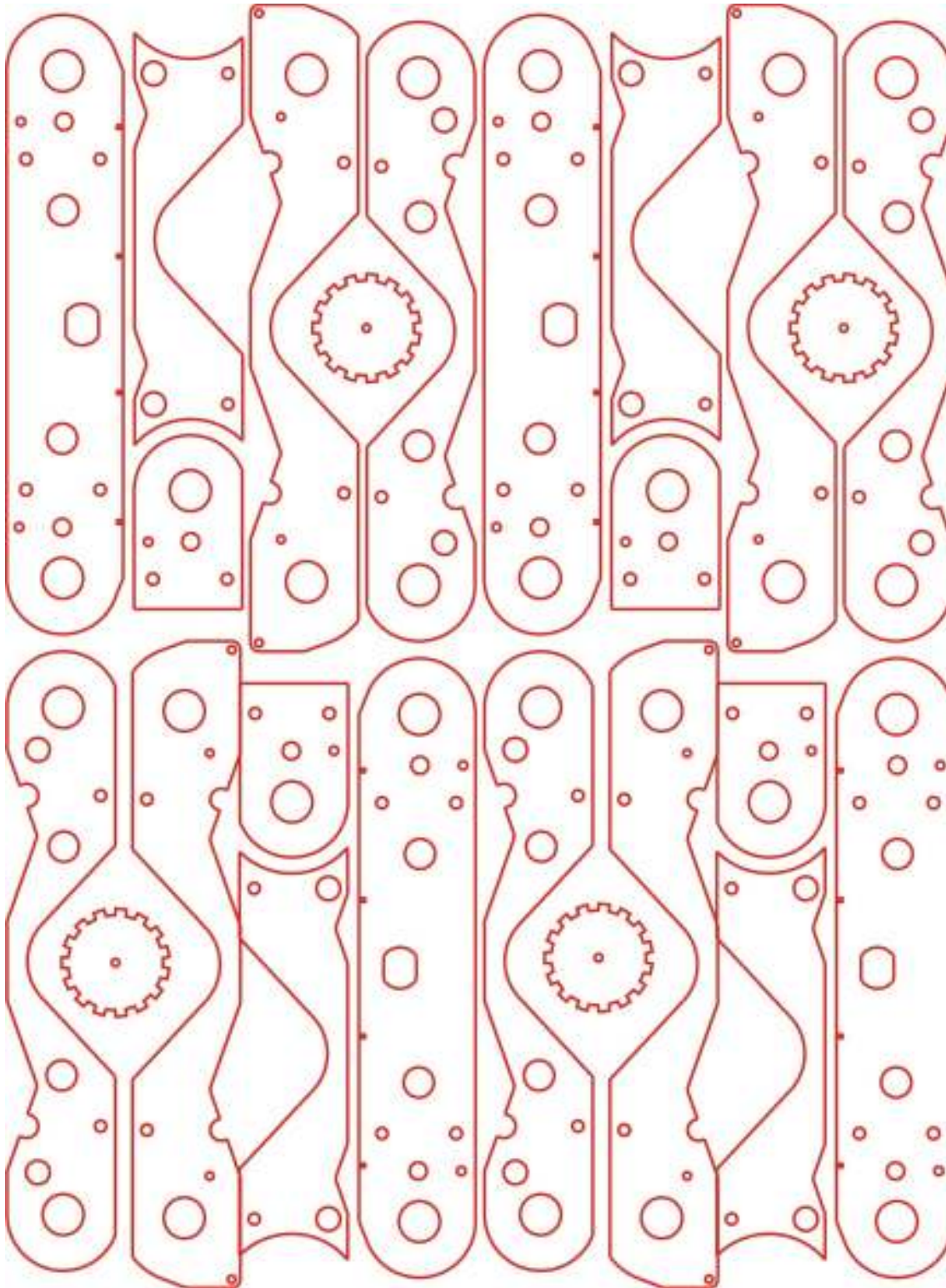


Figure A.2- Caterpillar frame mechanical schematics

A-3. Links and Gears

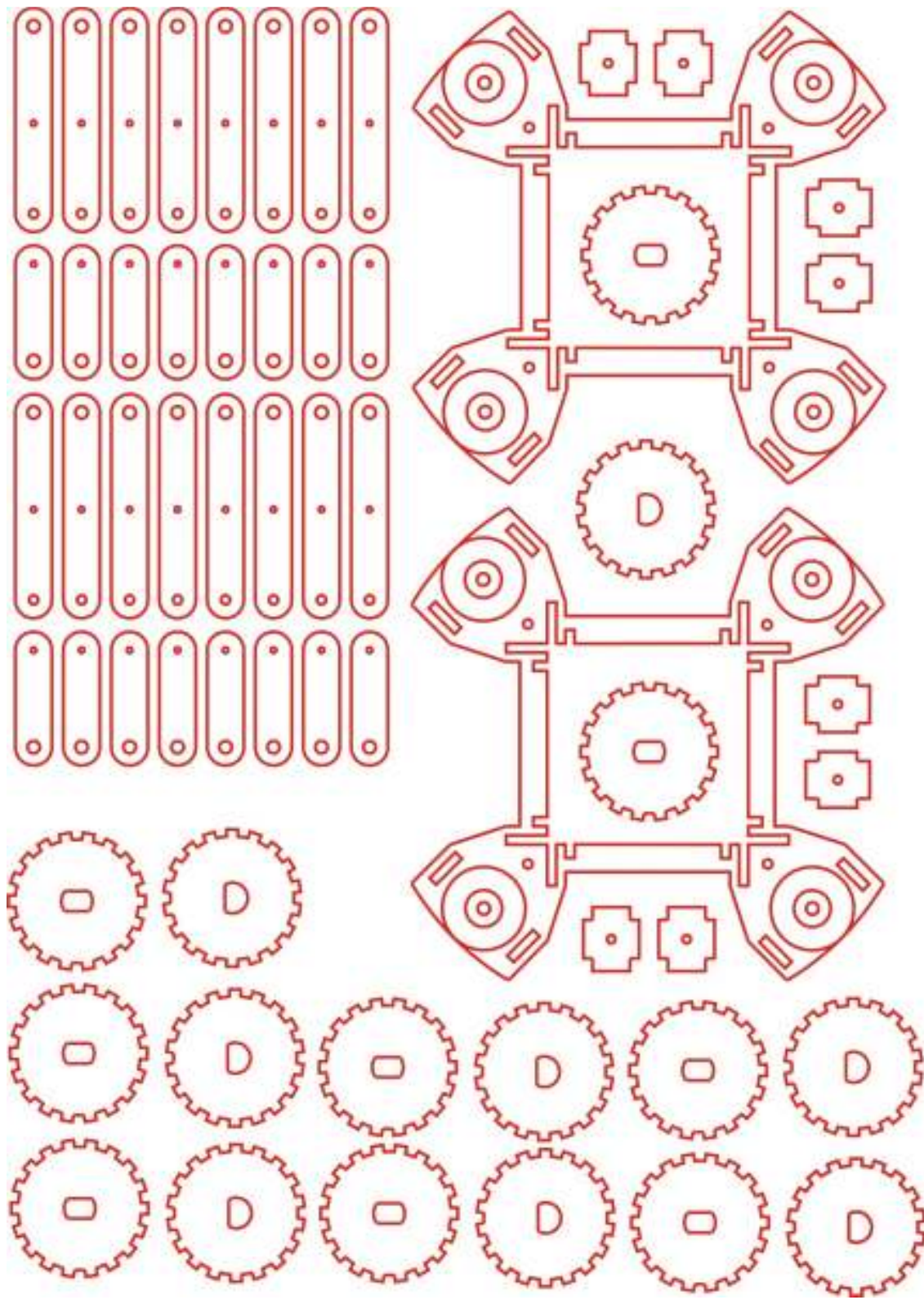


Figure A.3- Link and gear mechanical schematics

B. Electrical Schematics of Interface board

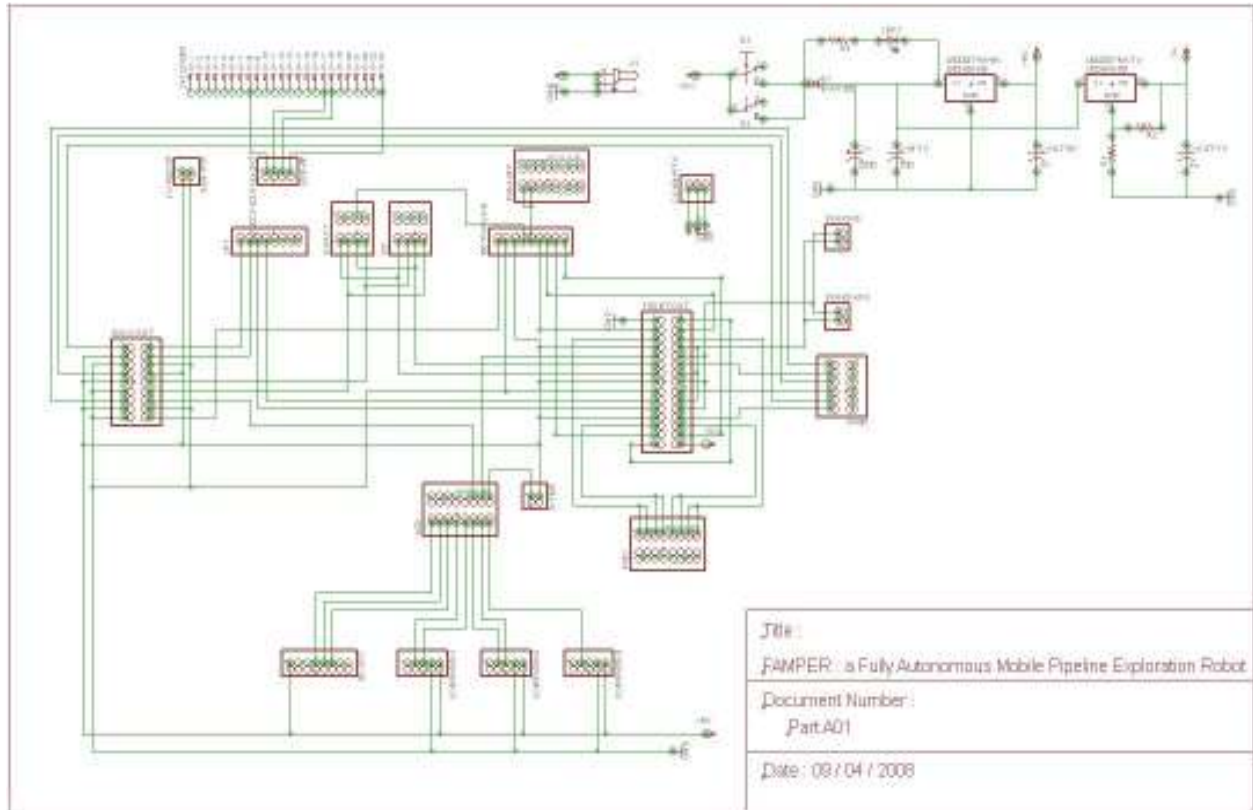


Figure A.4 - Electrical schematics of interface board

C. PCB of Interface Board

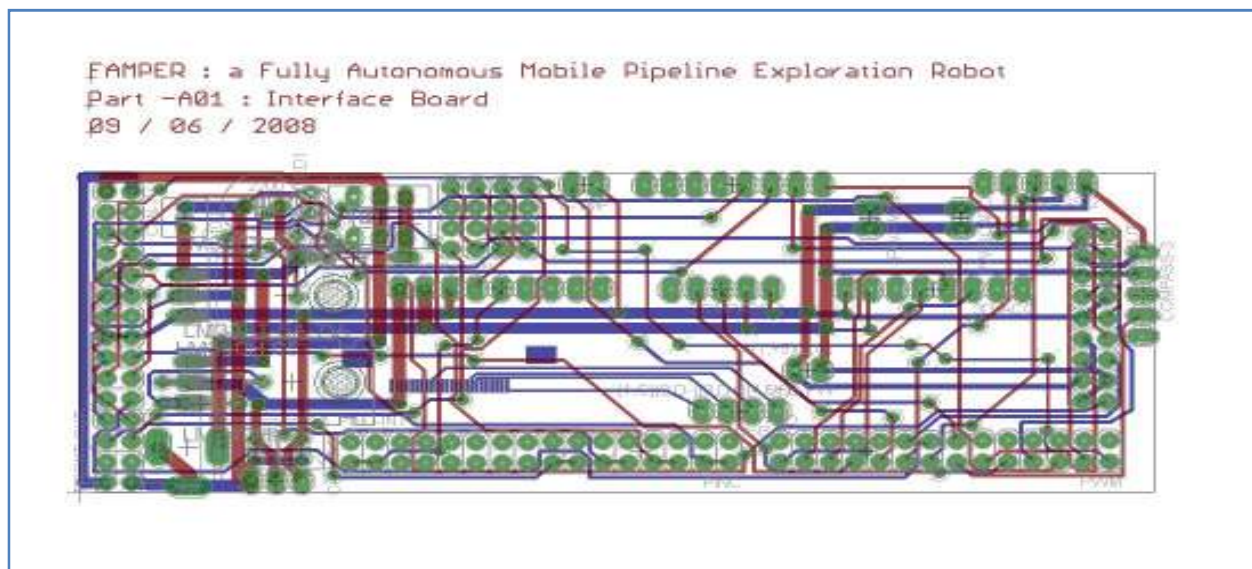


Figure A.5 - PCB of interface board

D. Detailed Gumstix Motherboard Specification

Table A.1 - Detailed Gumstix motherboard specification

Motherboard	Connex 400xm	Verdex XL6P
Processor	Marvell® PXA255 with XScale™	Marvell® PXA270 with XScale™
Speed	400 MHz	600 MHz
SDRAM memory	64 MB	128 MB
Flash Memory	16 MB	32 MB
Modular connectors	60-pin Hirose connector 92-pin bus header	60-pin Hirose connector 120-pin MOLEX connector 24-pin flex ribbon
Features	USB client signals 3 logic level UARTs NSSP DMA access I ² C™ Bus JTAG	USB Full Speed (v1.1) host signals CCD camera (CIF) signals 3 logic level UARTs NSSP DMA access Up to 98 GPIO's Power Management ³ I ² C™ Bus JTAG
Operating System	Linux® Buildroot	Linux® OpenEmbedded
Dimensions	80mm x 20mm x 5.9mm	80mm x 20mm x 6.3mm
Weight	8g (1/4 oz)	8g (1/4 oz)
Motherboard Temperature range	0°C to 85°C	-25°C to 85°C
Supply voltage	3.6V - 5.0V DC	3.6V - 5.0V DC
Current draw max.	~220mA @ 4.5V	~300mA @ 5.0V

APPENDIX II : GUMSTIX AND ROBOSTIX BASICS

A. Gumstix Basics

Gumstix is entirely based on the software framework called Open Embedded or Buildroot, which are the forms of Linux frameworks for embedded architectures. We can cross compile the packages needed for the project and port them to Gumstix using Serial interface, USB, MMC, Ethernet or Wireless mediums. Also the base Linux open embedded image needs to be built on a regular computer (with Linux preferred) by bitbaking the required recipe, which can be found at <http://www.gumstix.net>. The earlier way of building the Linux image for the Gumstix was done using Buildroot method of building images, which uses make files instead of bitbake recipes in the case of the more recently adapted procedure of building Linux images for Gumstix's.



Figure B.1 - Front and back view of Gumstix

As you might be wondering on how the packages compiled on one architecture work on an entirely different architecture. They won't work if we have compiled the packages for that particular machine's architecture, then we should be using a compiler called `avr-gcc` which is a gcc compiler for AVR architecture based chips. Thus, we could be sure that the packages will be compiled for AVR architecture. So as we will be using the `avr-gcc` to compile the packages, we can just transfer the binaries to the Gumstix by packaging all the binaries together. We will not

have the complete libraries on the Gumstix as we have minimal flash memory on the Gumstix. As even with the most recent Verdex based Gumstix motherboard, we will only have a flash of about 32 MB.

The basic process outline of getting a Gumstix to work for our needs can be outlined as, obtaining a buildroot, creating filesystem image, recompile the Linux kernel, replacing the filesystem image, cross-compiling any other required packages and transferring the binaries onto the Gumstix.

We will discuss the procedures of building the Linux image for the Gumstix as some of the packages that we need to compile for the Gumstix may not come with the bitbake recipe. In such a case we can still compile the package using the traditional makefile compilation procedure. Let us see how we can setup the environment needed for building the Linux image for Gumstix using either Buildroot procedure or the Open Embedded (bitbake based) procedure. We have used Ubuntu (Debian based Linux Distribution) as the base platform for setting up the Buildroot as well as open embedded environment.

After setting up the basic Ubuntu installation, we need to install some basic packages so that we will be able to get the flash filesystem image using Buildroot procedure. The packages required are, autoconf, automake, libtool etc. All of these packages, which are essential for setting up the build environment, can be installed using a single package in Ubuntu called build-essentials. Now we need some more packages depending on the way we would like to setup the flash filesystem image. For a person who prefers much like a graphical interface, we might also need to install a package called ncurses, which is provided as libncurses, libncurses-dev in Ubuntu. In addition we also need packages like bison, flex, texinfo, and set of utilities which will

provide the framework for generating multi-lingual messages; also some libraries which are used by these packages. All of these can be installed a single command in Ubuntu.

```
sudo apt-get install build-essential libncurses-dev bison flex texinfo zlib1g-dev gettext openssl-dev autoconf
```

Once we install all of the above packages, we then have to build environment setup. Now we need to check out the Buildroot packages from the Gumstix repository using svn and wget. Thus, in order to use them we need install subversion and wget on the machine, which can be done as,

```
sudo apt-get install subversion wget
```

And also once we have our flash filesystem image built, we need some kind of interface which will basically allow us to transfer files to the Gumstix and see what's happening with the Gumstix. In order to do so, we need some more packages like tftpd, kermit or minicom and lrzsz (serial z-modem drivers). We can install them as,

```
sudo apt-get install ckermit tftpd-hpa lrzsz
```

Finally we have completed setting up the basic environment needed for building the basic flash filesystem image. Now we are ready to check out the buildroot packages from the Gumstix repository.

```
svn co -rXXX http://svn.gumstix.com/gumstix-buildroot/trunk gumstix-buildroot
```

Here, XXX denotes the release number of the Buildroot which we would like to check out. So far, we have found out that versions 1161, 1541 and 1588 are the stables ones. So we can check out one of these releases. The more recent the release is, the more recent the kernel version is. So we might need to make sure all the packages that we will need for the setup of the Gumstix are indeed supported for that particular kernel version. Once we check out the Buildroot package

from the Gumstix repository, we can start the compilation by first creating the configuration file, which will indeed tell the makefile how to compile the packages and what to compile.

```
cd gumstix-buildroot  
rm .config  
make defconfig  
make
```

Here we get into the gumstix-buildroot folder which we checked out using svn, remove any pre-existing configuration file (if exists) and then make the default configuration file with the filename .config. Now once we get the .config file we can open this file using any text editor and edit the file by commenting out the packages which we do not need, and by un-commenting the packages which we need. If you think we need a better looking interface for editing the .config file, we can do,

```
make menuconfig
```

The menuconfig argument to the make command will give us a curses based interface for editing .config file. Once we have completed the making the configuration file, we can issue the make command to compile all the selected packages and make flash filesystem image. Once the flash filesystem image has been successfully built, we can find the image file with .jffs2 extension.

In order to use the latest flash file system image which we have built using the Buildroot package, we need to deploy that image onto the Gumstix. The flash image can be found in the gumstix-buildroot directory, which we checked out from the Gumstix repository and started the compilation in. Along with the rootfs.arm_nofpu.jffs2 file we can also find a couple of more files named uImage and u-boot.bin file. If you notice the directory structure of which the image was

made (Directory structure can be found under \$GUMROOT/build_arm_nofpu/root/), you can see that the /boot directory is missing in that structure. The boot directory has been packaged as another image ulmage, and u-boot.bin is the boot-loader file which tells the Gumstix where to look for the newly installed file system image.

For transferring the images which we have built till now, we need to connect the Gumstix to the computer which has those images using kermit or minicom. We will be using the serial connection to connect the Gumstix to the computer, so we can launch the kermit as,

```
kermit -l /dev/ttyS0
```

Here ttyS0 is the name given for the serial port on the Linux machine. Once we get connected to /dev/ttyS0, in order to listen to the Gumstix, we need to set the parameters on the kermit as required by the Gumstix.

```
set carrier-watch off  
set reliable  
fast  
set prefixing all  
set file type bin  
set rec pack 4096  
set send pack 4096  
set window 5  
set modem type none  
set flow-control none
```

Some of these parameters need to be set only if we are transferring files to the Gumstix (rec pack, send pack, window, modem type) as we are now. Once we set these parameters on the

kermit, we can simply start listening by typing 'connect' on kermit prompt. We can then start spewing out the output from the Gumstix, and stop at the login prompt. We can login to the Gumstix using the user name 'root' and the password 'gumstix'. The reason we are able to login is because the Gumstix is pre-installed with Linux from the factory. In order to change the file system image we need to stop the Gumstix at the boot-loader prompt 'GUM>'. We can do that by hitting any key while the Gumstix is starting and gives us a timeout of 3 seconds to get into the bootloader. Once we are at the boot-loader prompt we can start the transfer by issuing these series of commands.

```
GUM>loadb a2000000
kermit>send rootfs.arm_nofpu.jffs2
kermit>connect
GUM>pro on 1:0-1 && jera all && cp.b a2000000 40000 ${filesize}
```

Initially by loadb we are asking the Gumstix to load whatever comes next has to be received in binary mode at the given address. Then we go back to the kermit (by holding Ctrl + \ and c) and send the root file system image. Once we are done sending the file we connect back to the Gumstix and copy the image to the right location. And, we can transfer the uImage file in the similar fashion,

```
GUM>loadb a2000000
kermit>send uImage
kermit>connect
GUM>katinstall 100000
GUM>katload 100000
GUM>bootm
```

Once we complete transferring we need to install and load the kernel from its actual location. Now we will be able to boot into the newly installed flash file system image. We can go around the file system after logging into the Gumstix. Occasionally we might also want to flash the boot-loader so that we will be equipped with much more advanced commands from the boot-loader. Flashing the boot-loader with the new file u-boot.bin, which we obtained as a result of the Buildroot compilation, is in fact a process which has to be done with a lot of care, as a little mistake during this process can end up making the Gumstix dead. To recover the dead gumstix we might need to send it to the factory. The u-boot can be upgraded as,

```
GUM>loadb a2000000
kermit>send u-boot.bin
kermit>connect
GUM>protect off 1:0-1
GUM>era 1:0-1
GUM>cp.b a2000000 0 ${filesize}
GUM>reset
```

Basically everything here is similar to what we have been doing so far for replacing the images, but the only key difference is the destination address is 0 when we are replacing the u-boot image and its 40000 when we are replacing the jffs2 image.

In this thesis, we are also using another module called Robostix which operate sensors and motor-controllers. In order to interact between the Robostix and the Gumstix, we need to initialize all the modules for the Robostix and the utilities. We can enable some of these while building the buildroot image for the Gumstix, by making the selection for uisp. Now all we need is the Robostix module for the gumstix. We can build the Robostix modules for the Buildroot

version as,

```
cd (to the directory containing gumstix-buildroot)

svn co -rXXX http://svn.gumstix.com/gumstix-buildroot/branches/projects/robostix robostix

cd robostix/gumstix

make
```

Here it is important that we check out the Robostix repository in parallel to the Gumstix repository, in other words, both the gumstix-buildroot directory and the robostix directory in the same parent directory. Once we complete the compilation successfully, we can find the Robostix module as robostix_drv.ko. We need to copy all the essential files like, robostix_drv.ko, robostix, sertest onto the Gumstix. When we have these files on the Gumstix, we can insert the Robostix module by issuing the command,

```
insmod robostix_drv.ko (if this file is in the present working directory)

lsmod
```

Once we insert the module we can check if the Gumstix is recognizing the Robostix by issuing the command uisp, which will if everything is done successfully, acknowledge by saying “ATMega 128 found”.

B. Robostix Basics

Robostix is one of the expansion modules that is designed to accompany the Gumstix in applications. Robostix comes with an AVR ATmega 128 micro controller with the capability of converting analog to digital, digital to digital, and pulse width modulation. It has 8 ADC (Analog to Digital Converter) channels which will make it an eligible choice for this implementation of FAMPER. The Robostix board also consists of I/O pins that can be used to control any of the external sensors connected to them or modules that we might connect to the Robostix. For

instance we can connect additional LED's to the I/O pins of the Robostix and make them to dictate the status of the robot. Robostix can run the code as a standalone micro controller, but it has a very slower processor and smaller memory than the Gumstix. In order to use the sensor values in the Gumstix, we need to connect the Gumstix to the Robostix. Hence we can clearly say that Robostix will add much more flexibility to the Gumstix in every way.

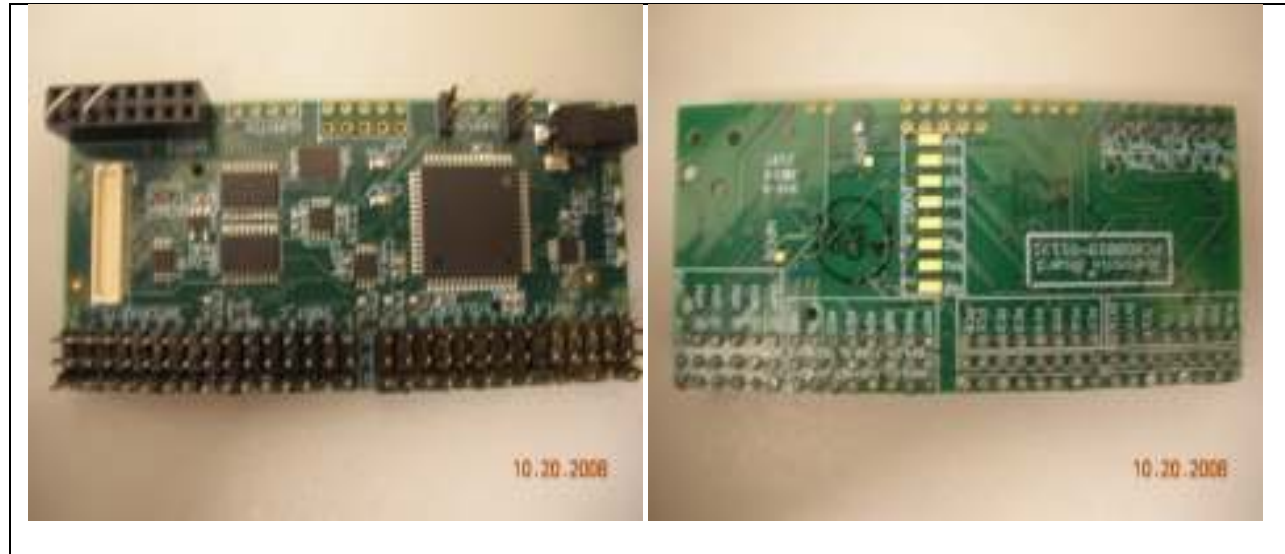


Figure 1 Front and Rear view of Robostix

As of now the Gumstix does not ship with the ability of converting analog to digital, but we can use Robostix which is included with an analog to digital converter and grab the digital outputs. Robostix inherits this ADC functionality from its AVR Atmega 128 micro controller. The Gumstix and the Robostix are connected together using serial port also called, STUART (/dev/ttyS2) and UART0, in other words the serial port STUART on Gumstix is connected to the UART0 serial port of Robostix in such a way that the Tx port of the Gumstix connected to the Rx port of the Robostix and the Rx port of the Gumstix connected to the Tx port of Robostix. Now as the Gumstix is now successfully connected to the Robostix we can use programs like uisp, which allows you to program the AVR Atmega 128 micro controller through the serial ports of the Gumstix and Robostix.

Table B.1 - UART

Function	Pin	Pin	Function
STUART RxD (GPIO 46)	1	2	Gnd
+5	3	4	robostix UART-0 TxD (Port E.1)
STUART TxD (GPIO 47)	5	6	+5
Gnd	7	8	robostix UART-0 RxD (Port E.0)
robostix UART-1 RxD (Port D.2)	9	10	Gnd
+5	11	12	I2C SDA (Port D.1) labelled TX
robostix UART-1 TxD (Port D.3)	13	14	+5
Gnd	15	16	I2C SCL (Port D.0) labelled RX

Table B.2 -. Analog to digital (A/D)

Pin	Label	GPIO	JTAG
1	AD0	Port F.0	
2	AD1	Port F.1	
3	AD2	Port F.2	
4	AD3	Port F.3	
5	AD4	Port F.4	TCK
6	AD5	Port F.5	TMS
7	AD6	Port F.6	TDO
8	AD7	Port F.7	TDI

The communication protocol on the Robostix waits for any incoming ASCII characters on UART0, that would represent ADC channel numbers from 0 to 7, and then responds with the current voltage value on that particular ADC channel, in hexadecimal format encoded as ASCII

string. As we have a little bit of basics on how the Robostix works and all, we can start and learn on how to flash new programs onto the Robostix.

Table B.3 - PortC of Robostix

Pin	Label	GPIO
1	PC0	Port C.0
2	PC1	Port C.1
3	PC2	Port C.2
4	PC3	Port C.3
5	PC4	Port C.4
6	PC5	Port C.5
7	PC6	Port C.6
8	PC7	Port C.7

We can compile the program that we have made for the Robostix using a makefile that will generate the hex file for flashing the Robostix. Once we have the required hex file on our machine, we can connect to the Gumstix and insert the Robostix module so that the Gumstix could see the Robostix.

```
insmod robostix_drv.ko  
  
lsmod
```

Once the Robostix module is inserted, we can start the application uisp by using the command,

```
# uisp  
  
AVR ATMega 128 Found
```

if the uisp program does not acknowledge with the AVR ATMega found message, you might need to check if you have the connected the Rx port of the STUART to Tx of UART0 and Tx of

STUART to Rx of UART0. Now once we get the acknowledgment, we need to set the fuses on the Robostix. We can do that by,

```
uisp --wr_fuse_l=0xbf --wr_fuse_h=0xc9 --wr_fuse_e=0xff
```

Now the Robostix is ready to accept the flash file for the micro controller. We can send the hex file to the Robostix using the command,

```
uisp --erase --upload if=Motor.hex
```

With the added capabilities of Robostix on Gumstix, the FAMPER's architecture will certainly give the ability of being a Fully Autonomous Mobile robot.

VITA

Jong-Hoon Kim was born in Jeongseon, South Korea, in November 1972. He graduated Jaedong Elementary School in Seoul, South Korea, followed by Choongang Middle School in the same city. Next, he graduated Jeongseon High School in Jeongseon, South Korea, in February 1991. He served in Korean Army for twenty seven months. In February 1996, he earned a college diploma from department of legal studies at KyungBuk College, South Korea. He was employed as a system administrator for NICES project in IBM-Korea from April 2000 to February 2002. He worked as a system supervising manager in I-BUS from March 2002 to September 2004. During his tenure at I-BUS, he earned a Bachelor of Engineering in Seoul National University of Technology in Seoul, South Korea, 2005. Further, he began his studies at Louisiana State University in August of 2005. JongHoon Kim will receive the degree of Master of Science in System Science in December 2008.